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Questions

Q1.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

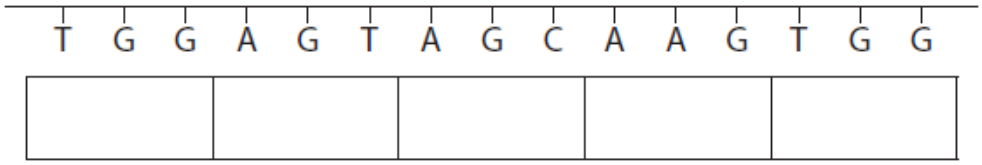
The sequence of bases in DNA determines the sequence of amino acids in a polypeptide.

The table shows four amino acids and their genetic codes.

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Amino acid	Genetic code
alanine (Ala)	GCT or GCC or GCA or GCG
lysine (Lys)	AAA or AAG
serine (Ser)	AGT or AGC or TCT or TCC or TCA or TCG
tryptophan (Trp)	TGG

(a) The diagram shows a DNA base sequence.



(i) Complete the diagram to show the sequence of amino acids coded by this DNA base sequence.

(1)

(ii) Explain why only five amino acids are coded by this sequence of bases.

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(2)

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(b) Explain why some amino acids, such as alanine, have more than one genetic code.

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(3)

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(c) Of the 64 possible genetic codes, 61 code for amino acids.

(i) Calculate the percentage of genetic codes that code for amino acids.

Give your answer to four significant figures.

(1)

Answer ..... %

(ii) Explain the role of the other three genetic codes.

(2)

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(d) Messenger RNA (mRNA) and transfer RNA (tRNA) are involved in the synthesis of a polypeptide chain from DNA.

Which row of the table shows the codon on mRNA and the anticodon on tRNA that correspond to tryptophan?

(1)

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	Codon on mRNA	Anticodon on tRNA
☒ A	ACC	TGG
☒ B	ACC	UGG
☒ C	UCC	AGG
☒ D	UCC	TGG

(Total for question = 10 marks)

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Q2.

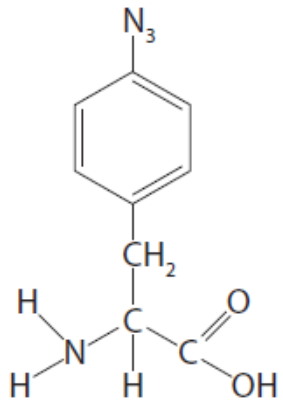
Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Silkworms are caterpillars that produce silk. Silk is a fibrous protein that can be used in clothing and in medicine.

Silkworms have been selectively bred to produce a modified silk that could have even more medical uses.

The modified silk is made by these silkworms by inserting a synthetic amino acid, AzPhe, into the protein. This replaces the naturally-occurring amino acid phenylalanine.

The diagram shows the structure of AzPhe.



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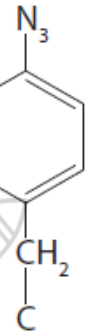
(a) Which of the following diagrams shows the R group only of AzPhe?

(1)

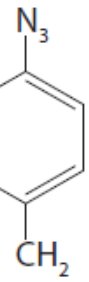
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☒ **A**

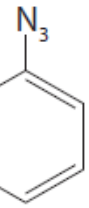


☒ **B**

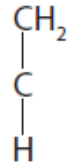


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☒ **C**



☒ **D**



(b) Only the silkworms that have been selectively bred can use AzPhe in the synthesis of protein.

During translation, AzPhe joins to two other amino acids by condensation reactions.

(i) Name the covalent bond that joins two amino acids in a condensation reaction.

(1)

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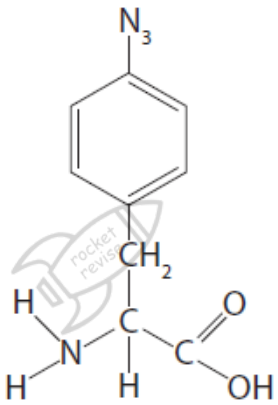
(ii) Draw two circles on the diagram of AzPhe to show which parts of the molecule

are lost when AzPhe joins to two other amino acids by condensation reactions.

(2)

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(iii) Transfer RNA (tRNA) is involved in translation. The amino acid AzPhe requires a special tRNA molecule during the synthesis of silk.

Suggest why AzPhe is **not** inserted into proteins in silkworms that have not been selectively bred.

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(3)

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(c) Part of a silk molecule contains 1100 amino acids. In natural silk, 1% of the amino acids are phenylalanine.

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In modified silk, 16% of the phenylalanine molecules are replaced by AzPhe.

(i) Calculate the ratio of phenylalanine to AzPhe in this part of a modified silk molecule.

(2)

Answer .....

(ii) The R group of phenylalanine is smaller than the R group of AzPhe.

Suggest how inserting an amino acid with a larger R group could affect the properties of silk fibres.

(4)

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(Total for question = 13 marks)

Q3.

Meselson and Stahl performed experiments that demonstrated semi-conservative replication of DNA.

(a) (i) State what is meant by the term semi-conservative replication.

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(2)

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(ii) Name **one** enzyme involved in semi-conservative replication.

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(1)

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(iii) Explain the importance of semi-conservative replication in the production of new cells.

(2)

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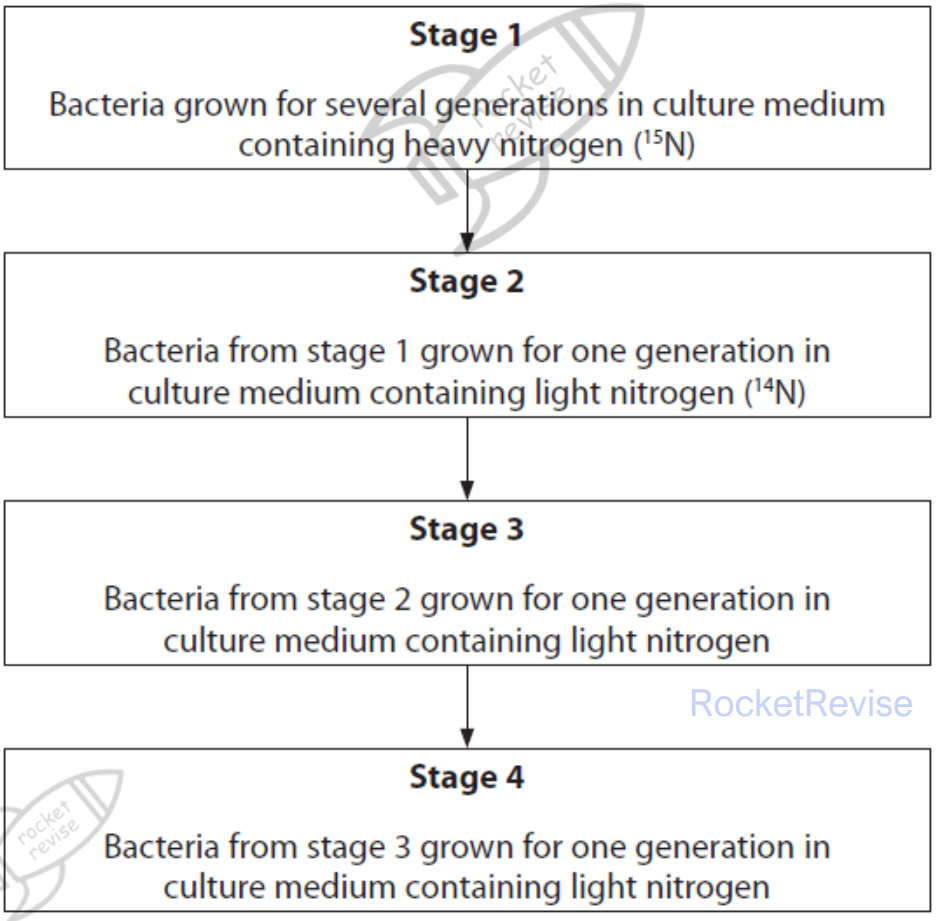
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(b) The flow chart summarises part of one experiment performed by Meselson and Stahl.



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Complete the table to show the percentage of the total number of DNA molecules containing heavy nitrogen only, light nitrogen only or both heavy and light nitrogen, at the end of each stage.

(3)

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End of stage	Percentage of DNA molecules containing		
	heavy nitrogen only	light nitrogen only	both heavy and light nitrogen
1			
2			
3			

(Total for question = 8 marks)

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Q4.

Nucleic acids include DNA and RNA.

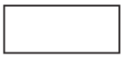
(a) Each single strand of a DNA molecule is synthesised from mononucleotides.

Draw a diagram to show two mononucleotides joined together in a single strand of DNA.

Use the symbols shown for each component in your diagram.

(3)

Phosphate group 

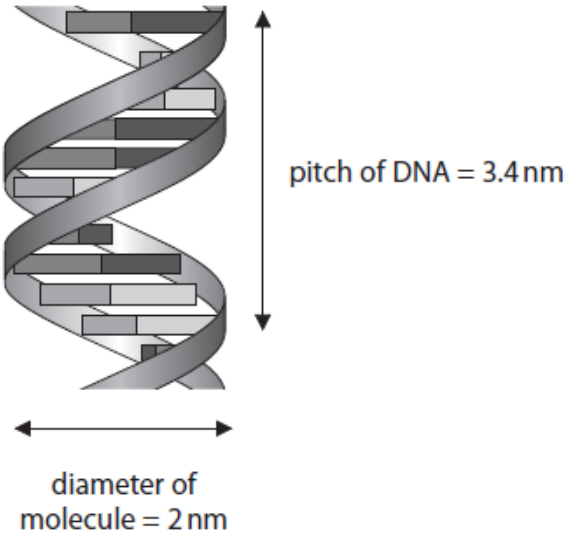
Base 

Pentose sugar 

Bond

(b) The diagram represents part of a DNA molecule.

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The pitch is the length of one complete turn in the double helix.

There are 10 base pairs in one pitch.

(i) Calculate the distance between one base and the next base on one strand of DNA.

Give your answer to an appropriate number of significant figures.

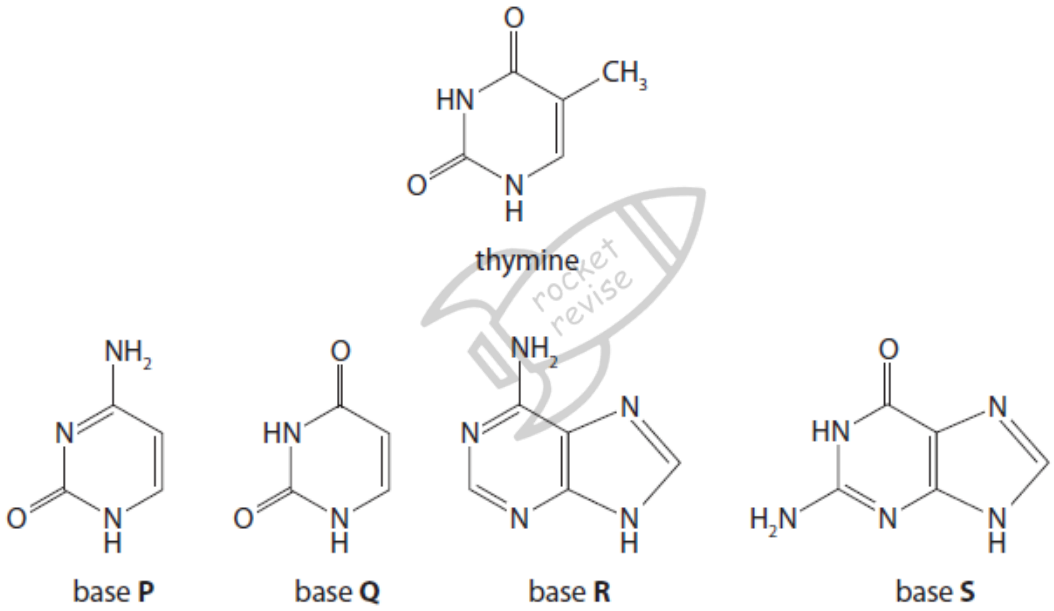
(1)

Answer ..... nm

(ii) The diagram shows the structure of thymine and four other bases, **P**, **Q**, **R** and **S**.

Bases **P** and **S** can form three hydrogen bonds each and bases **Q** and **R** can form two hydrogen bonds each.

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Explain which of the four bases **P**, **Q**, **R** or **S** is adenine.

(2)

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(c) Compare and contrast the structure of messenger RNA (mRNA) with the structure of transfer RNA (tRNA).

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(Total for question = 10 marks)

Q5.

Errors in DNA replication can give rise to mutations.

The diagram shows the bases in a length of DNA.

Length of DNA	A	T	G	C	T	C	A	T	T	T	A	C	C	A	T	C	G	A
Base number	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18

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The table shows the genetic code for the amino acids.

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Genetic code	Amino acid	Genetic code	Amino acid	Genetic code	Amino acid	Genetic code	Amino acid
AAA AAG	Lysine	CAA CAG	Glutamine	GAA GAG	Glutamic acid	TAC TAT	Tyrosine
AAC AAT	Asparagine	CAT CAC	Histidine	GAC GAT	Aspartic acid	TCA TCC TCG TCT	Serine
ACA ACC ACG ACT	Threonine	CCA CCC CCG CCT	Proline	GCA GCC GCG GCT	Alanine	TGG	Tryptophan
AGA AGG	Arginine	CGA CGC CGG CGT	Arginine	GGA GGC GGG GGT	Glycine	TGC TGT	Cysteine
AGC AGT	Serine	CTA CTC CTG CTT	Leucine	GTA GTC GTG GTT	Valine	TTA TTG	Leucine
ATA ATC ATT	Isoleucine					TTC TTT	Phenylalanine
ATG	Methionine						

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The genetic codes TAA, TAG and TGA are stop codons.

(a) State the sequence of the first four amino acids coded for by this length of DNA.

(1)

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(b) A change in a single base can cause a change in the amino acid sequence produced in protein synthesis.

(i) Name the type of each mutation described below.

(2)

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Base number 3 becomes cytosine (C)

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Base number 6 becomes number 5 in the sequence

.....

Base number 9 becomes number 10 in the sequence

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*(ii) Explain the possible effects of these three types of mutation on the amino acid sequence coded for by this length of DNA.

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Use the information in the table to support your answer.

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(Total for question = 9 marks)

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Q6.

Warfarin is used in the treatment of cardiovascular disease.

Warfarin inhibits the synthesis of prothrombin.

(a) Describe the role of prothrombin in the blood clotting process.

(3)

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(b) Which type of treatment is warfarin?

(1)

☒ **A** anticoagulant

☒ **B** antihypertensive

☒ **C** platelet inhibitor

☒ **D** statin

(c) Reduced vitamin K is needed for the synthesis of prothrombin.

An enzyme, vitamin K epoxide reductase (VKOR), converts vitamin K to reduced vitamin K.

The diagram shows this conversion.

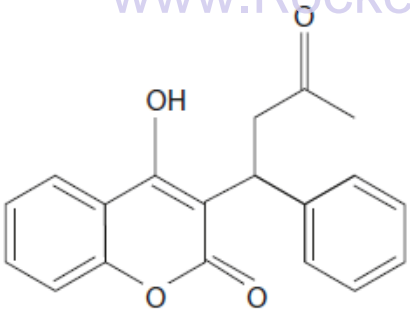
$$\text{vitamin K} \xrightarrow{\text{VKOR}} \text{reduced vitamin K}$$

(i) The diagrams show the structures of warfarin and vitamin K.

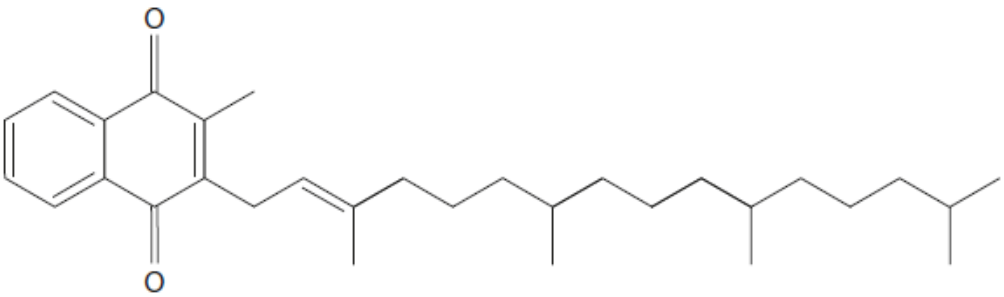
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Warfarin



Vitamin K

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Using the information in the diagrams, suggest why warfarin inhibits the synthesis of prothrombin.

(2)

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(ii) Suggest why people taking warfarin have to avoid eating foods that contain a high concentration of vitamin K.

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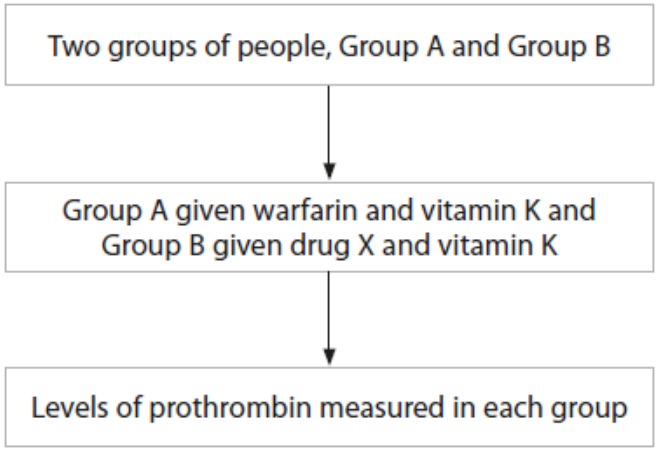
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(d) Other drugs are available that work in a similar way to warfarin.

The diagram shows one study to compare the effect of warfarin with drug X.



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Explain how this study should be designed so that the effectiveness of these two drugs can be compared.

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(Total for question = 12 marks)

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Q7.

A gene contains the genetic code for the sequence of amino acids in a polypeptide chain.

The table shows the genetic codes found in DNA.

Genetic code	Amino acid	Genetic code	Amino acid	Genetic code	Amino acid	Genetic code	Amino acid
AAA AAG	Lysine	CAA CAG	Glutamine	GAA GAG	Glutamic acid	TAC TAT	Tyrosine
AAC AAT	Asparagine	CAT CAC	Histidine	GAC GAT	Aspartate	TCA TCC TCG TCT	Serine
ACA ACC ACG ACT	Threonine	CCA CCC CCG CCT	Proline	GCA GCC GCG GCT	Alanine	TGG	Tryptophan
AGA AGG	Arginine	CGA CGC CGG CGT	Arginine	GGA GGC GGG GGT	Glycine	TGC TGT	Cysteine
AGC AGT	Serine	CTA CTC CTG CTT	Leucine	GTA GTC GTG GTT	Valine	TTA TTG	Leucine
ATA ATC ATT	Isoleucine					TTC TTT	Phenylalanine
ATG	Methionine						

The genetic codes TAA, TAG and TGA are stop codons, that do not code for amino acids.

*(a) Explain the nature of the genetic code.

Use information in the table to support your answer.

(6)

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(b) The diagram shows the sequence of nucleotide bases in part of a DNA template (antisense) strand.

Number	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
Base	A	T	G	G	C	T	T	G	C	C	C	G	A	T	C	C	T	A

(i) Give the sequence of amino acids that is coded for by these bases.

(1)

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(ii) Explain the possible effects on a protein if there is a substitution mutation in the 9th base in this DNA strand.

Use information in the table to support your answer.

(5)

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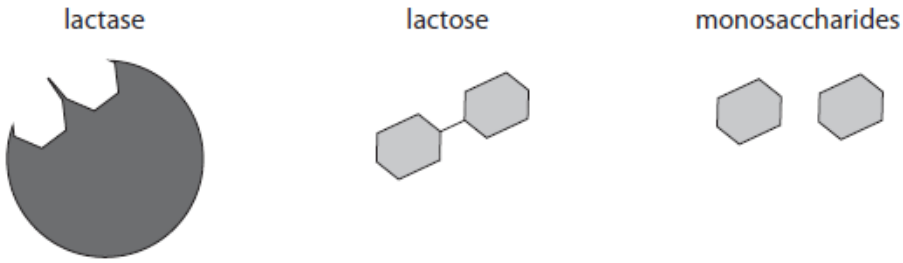
(Total for question = 12 marks)

Q8.

Answer the question with a cross in the boxes you think are correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Lactase is an enzyme that catalyses the hydrolysis of lactose.

(a) (i) The diagrams show symbols for lactase, lactose and the monosaccharides of lactose.



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Which diagram shows the sequence of events for the hydrolysis of lactose?

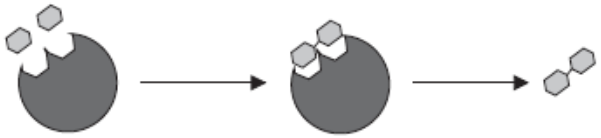
(1)

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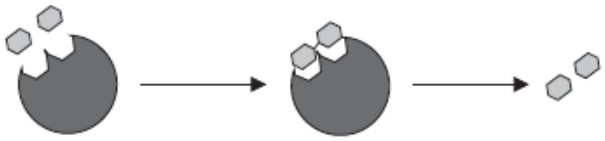
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☒ A 

☒ B 

☒ C 

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☒ D 

(ii) Which of the following are the monosaccharides present in lactose?

(1)

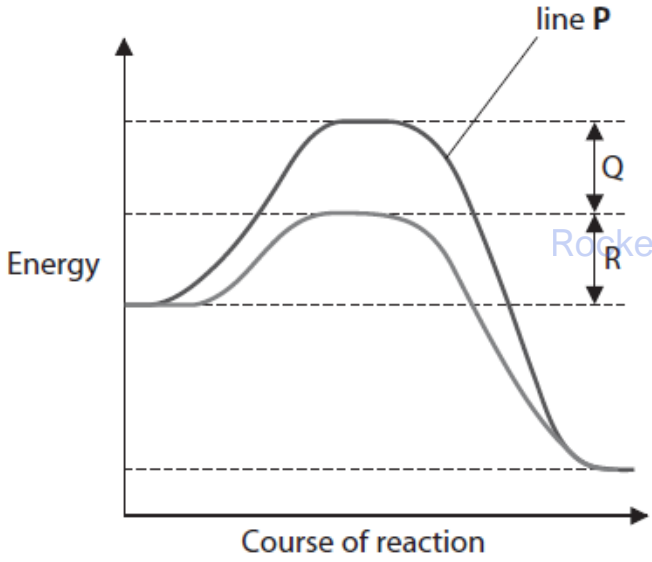
☒ **A** fructose and sucrose

☒ **B** galactose and glucose

☒ **C** glucose and fructose

☒ **D** sucrose and galactose

(b) The graph shows the energy changes for the hydrolysis of lactose, with and without lactase.



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Which row in the table describes line **P** and identifies the letter that shows the decrease in activation energy due to lactase?

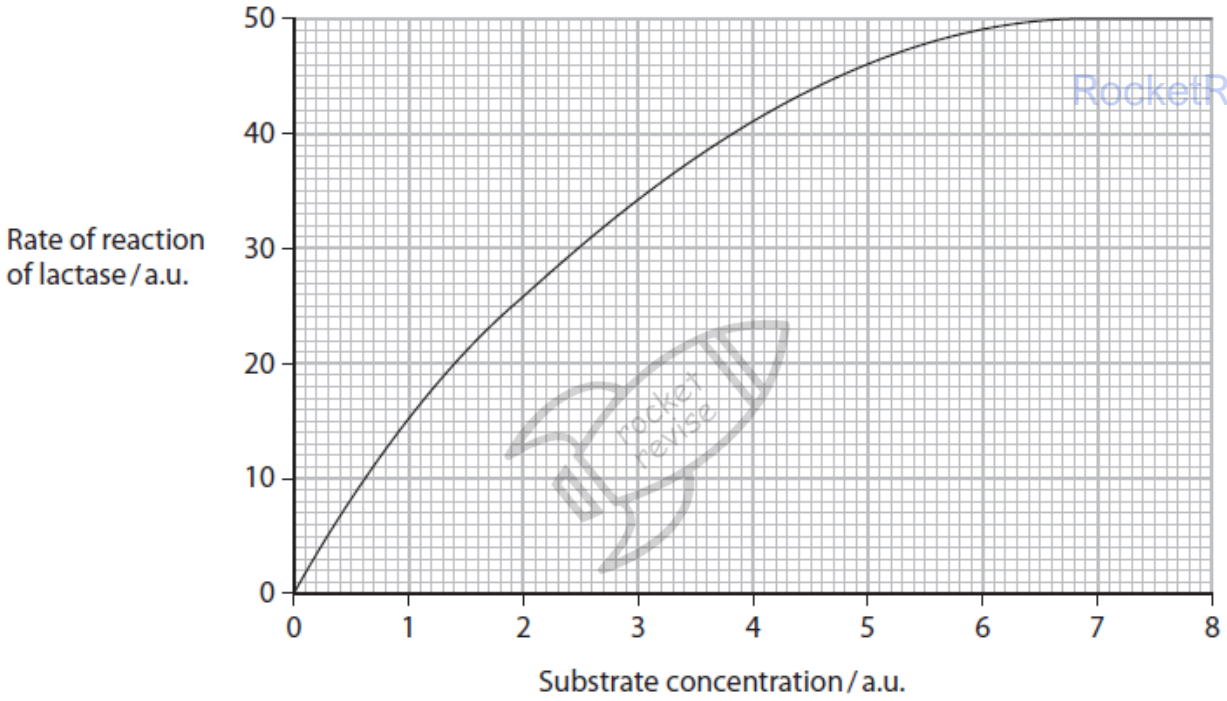
(1)

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	Line P	Decrease in activation energy due to lactase
☒ A	reaction with enzyme	Q
☒ B	reaction with enzyme	R
☒ C	reaction without enzyme	Q
☒ D	reaction without enzyme	R

(c) The graph shows the relationship between substrate concentration and the rate of reaction of lactase.



(i) Explain why substrate concentration affects the rate of reaction.

(2)

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(ii) The reaction rate (V) at each substrate concentration can be calculated using the formula

$$V = \frac{V_{max} \times S}{K + S}$$

V_{max} is the maximum rate of reaction

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K is the substrate concentration when the rate of reaction is half the rate of V^{max}

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S is the substrate concentration.

Calculate the reaction rate (V) at a substrate concentration of 4 a.u.

(2)

Answer .....

(Total for question = 7 marks)

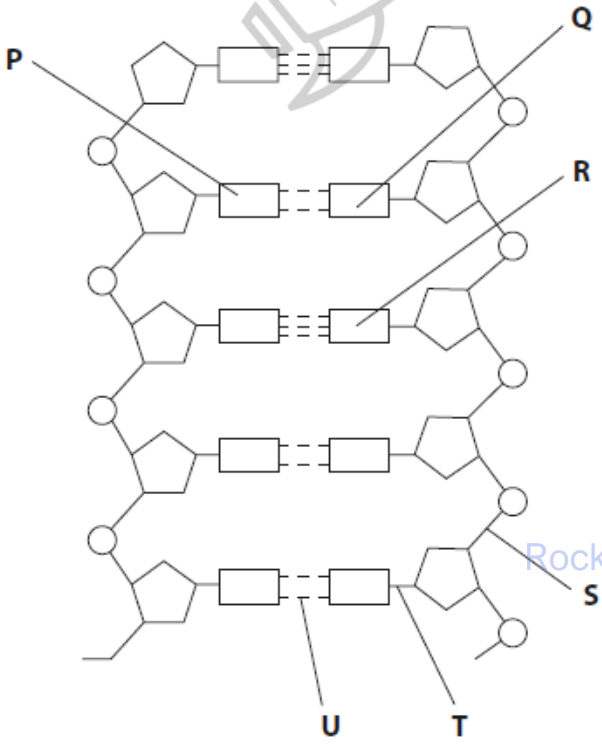
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Q9.

The polynucleotide DNA is composed of mononucleotides linked together.

Two polynucleotides form a DNA molecule.

(a) The diagram shows part of a DNA molecule.



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(i) Draw a circle around **one** mononucleotide that includes the base labelled **R**.

(1)

(ii) Which row of the table identifies the bonds labelled **S**, **T** and **U**?

(1)

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	S	T	U
☒ A	hydrogen	phosphodiester	covalent
☒ B	hydrogen	covalent	phosphodiester
☒ C	phosphodiester	hydrogen	covalent
☒ D	phosphodiester	covalent	hydrogen

(iii) The base labelled **P** is adenine.

Which is the base labelled **Q**?

(1)

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☒ **A** cytosine

☒ **B** guanine

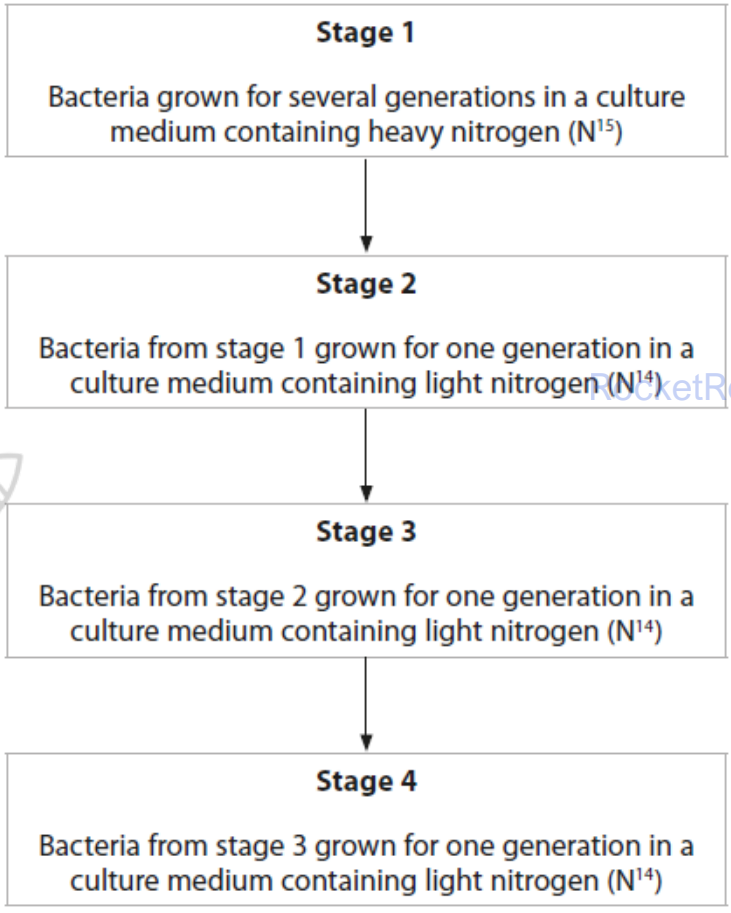
☒ **C** thymine

☒ **D** uracil

(b) Meselson and Stahl carried out experiments that provided evidence for the semi-conservative replication of DNA.

Heavy nitrogen (N^{15}) and light nitrogen (N^{14}) were used in these experiments.

The flow chart summarises part of one experiment performed by Meselson and Stahl.



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After each stage, a sample of DNA was taken from the bacteria and the DNA molecules separated using a density gradient in a tube.

The heavier DNA molecules form bands lower down the gradient than the lighter DNA molecules.

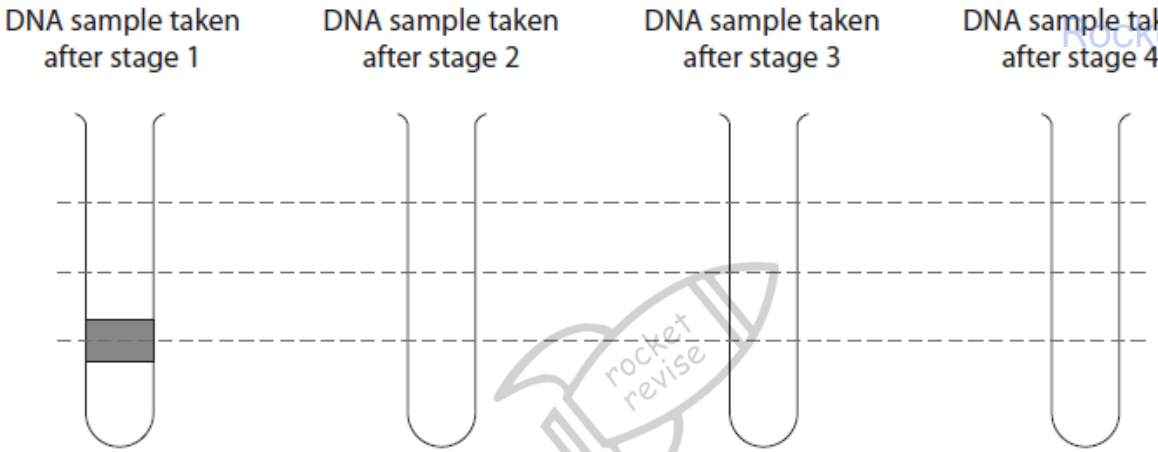
The height of each band is proportional to the percentage of DNA molecules in the sample.

(i) Complete the diagram to show the results of this experiment.

Use the dotted lines to help you to position the bands on the diagram.

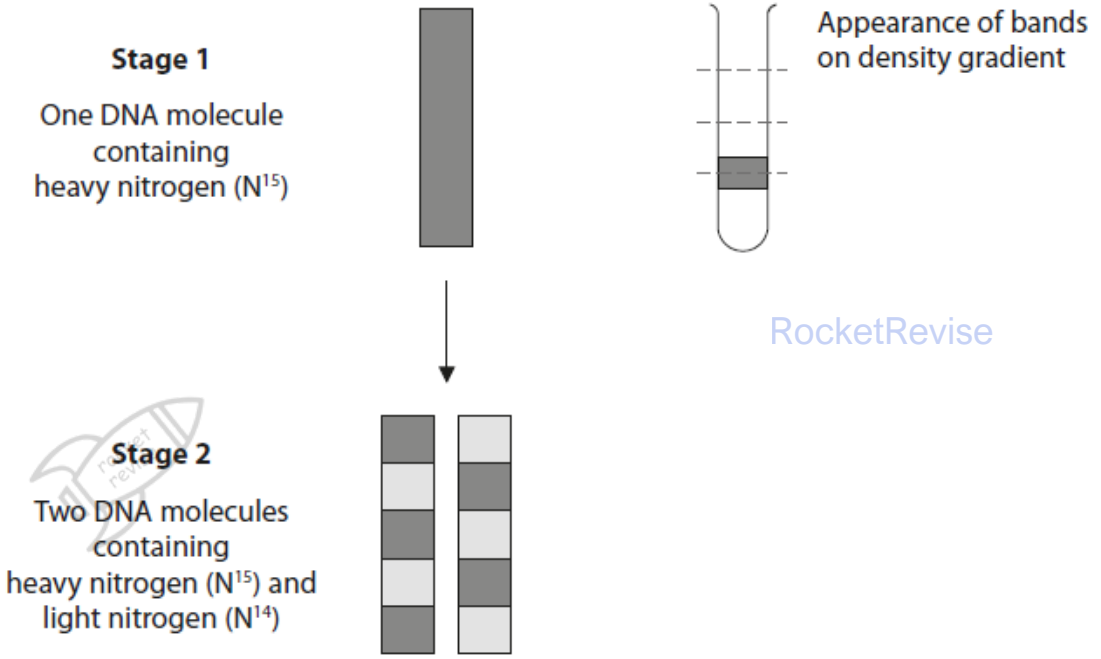
The first one has been done for you.

(5)



(ii) The experiments of Meselson and Stahl disproved the dispersive theory of DNA replication.

The diagram shows the expected results if the dispersive theory was correct.



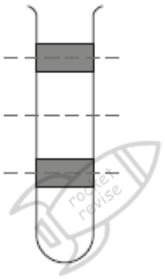
Which diagram would show the bands of DNA molecules on the density gradient at stage 2, if the dispersive theory was correct?

(1)

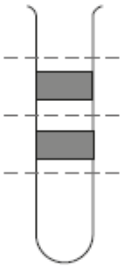
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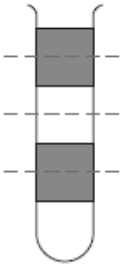
☒ A



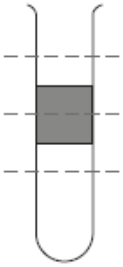
☒ B



☒ C



☒ D



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(Total for question = 9 marks)

Q10.

Polynucleotides include DNA and messenger RNA (mRNA).

Some components of polynucleotides are found in both DNA and mRNA.
Other components are found only in DNA or in mRNA.

A Venn diagram can be drawn to represent this information. Components found in both DNA and mRNA are written in the part of the diagram where the circles overlap.

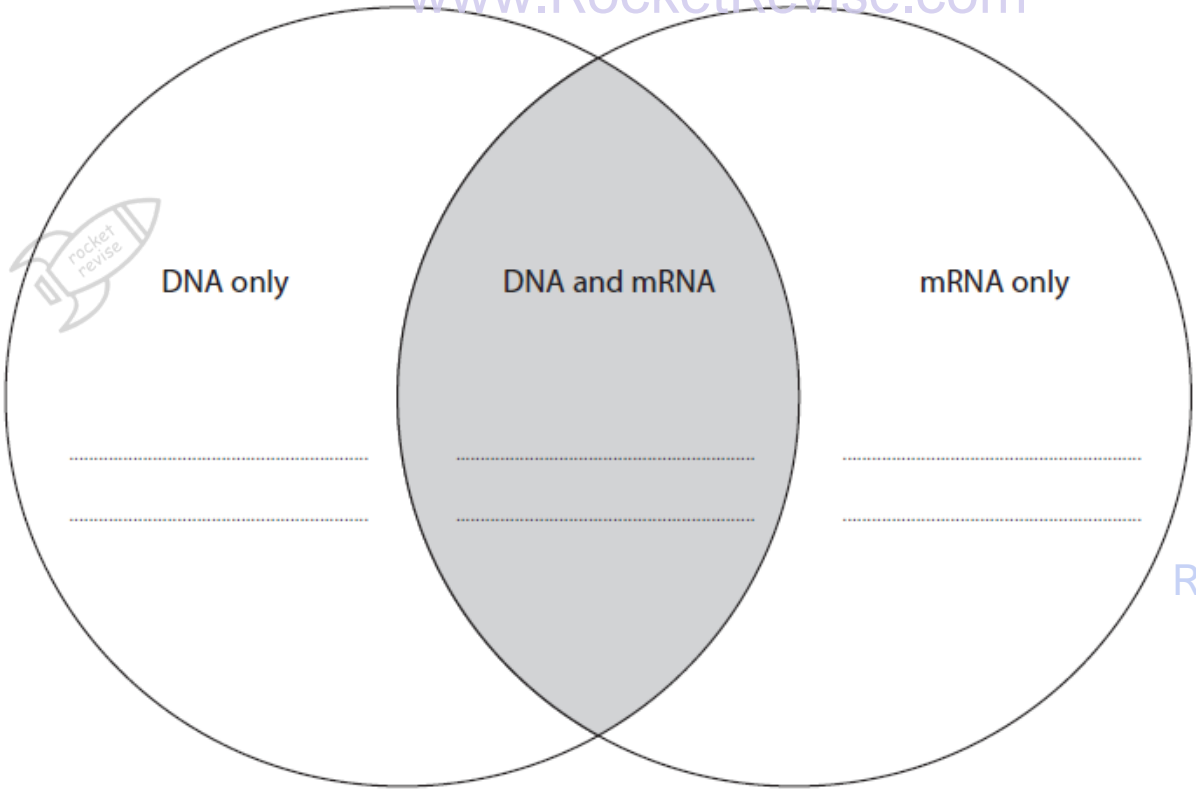
Complete the Venn diagram by writing the name of **two** components in each part of the diagram.

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(6)

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(Total for question = 6 marks)

Q11.

Answer the question with a cross in the box you think is correct ☒ . If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒ .

Warfarin is a drug used to treat people who have a blood clot.

(a) (i) Read through the following passage about warfarin.

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Write on the dotted lines the most appropriate word or words to complete the passage.

(3)

Warfarin is used to treat people with blood clots as it lowers the number of clotting factors in the blood.

One clotting factor in blood is prothrombin.

Prothrombin is converted to the enzyme thrombin by .....

The ..... of thrombin binds to fibrinogen and as a result a mesh of fibres and ..... is formed.

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(ii) Which type of drug is warfarin?

(1)

☒ **A** an anticoagulant

☒ **B** an antihypertensive

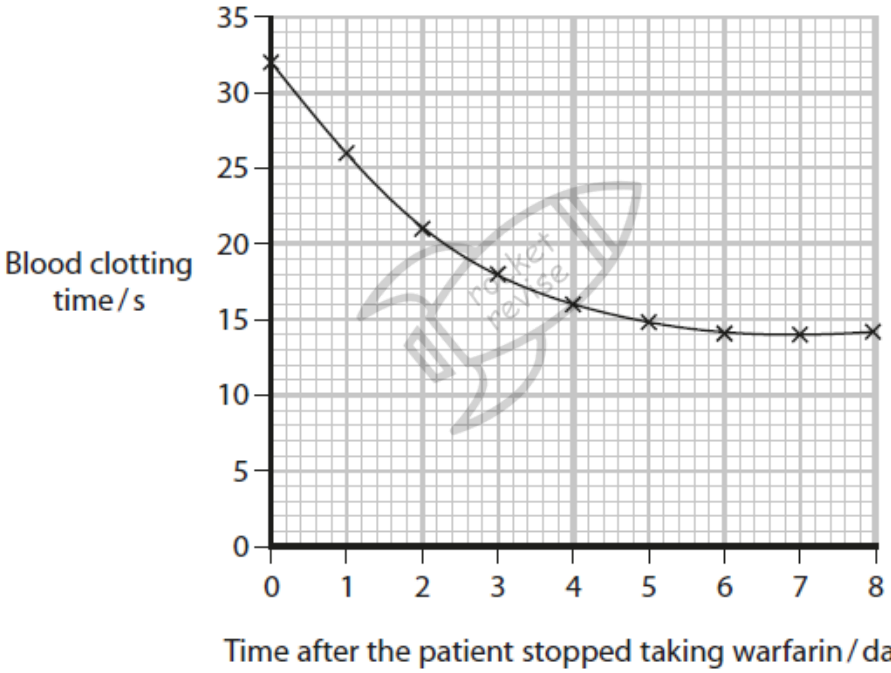
☒ **C** a platelet inhibitor

☒ **D** a statin

(b) The time taken for a blood sample to form a blood clot can be measured. This is called the clotting time.

The graph shows the blood clotting time after a patient has stopped taking warfarin.

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Calculate the rate of decrease in the clotting time at two days after stopping taking warfarin.

Use a tangent for your calculation.

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(2)

Answer .....

(Total for question = 6 marks)

Q12.

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Polysaccharides, lipids, nucleic acids and proteins are large molecules found in living organisms.

(a) Polymers are large molecules made of monomers. The table gives information about some of these polymers.

Complete the table by filling in the empty boxes with either the name of the monomer, the elements present in each monomer or the type of bond between monomers.

(4)

Polymer	Monomer	Elements present in monomer	Type of bond between monomers
polysaccharides	monosaccharide	.....	glycosidic
nucleic acid	.....	carbon, hydrogen, oxygen, phosphorus and nitrogen	.....
protein	amino acid	.....	peptide

(b) Triglycerides are lipids.

Describe how an unsaturated triglyceride is synthesised.

(3)

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(Total for question = 7 marks)

Q13.

Answer the question with a cross in the box you think is correct ☒ . If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒ .

Both DNA and RNA are polynucleotides.

(a) The table shows how the components of polynucleotides can be represented.

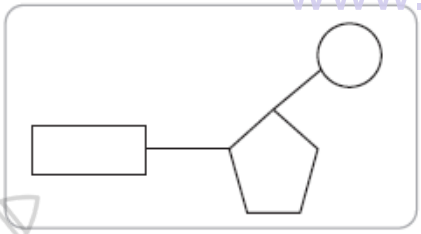
Component	Representation
sugar	
phosphate group	
cytosine or thymine	
adenine or guanine	
covalent bond	
hydrogen bond	

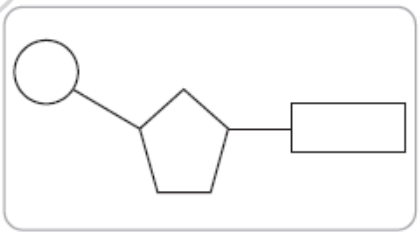
(i) Which diagram shows a mononucleotide?

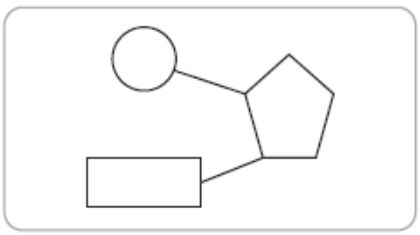
(1)

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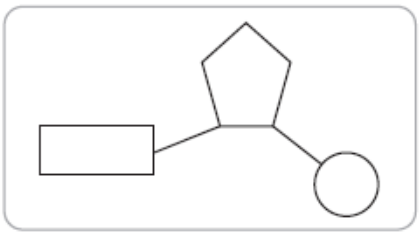
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☒ A 

☒ B 

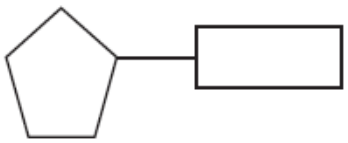
☒ C 

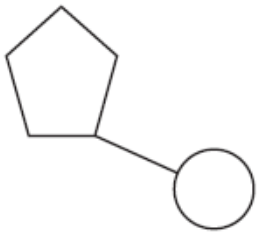
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☒ D 

(ii) Which diagram shows two components joined by a phosphodiester bond?

(1)

☒ A 

☒ B 

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☒ C 

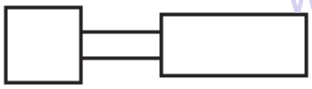
☒ D 

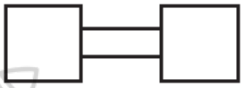
(iii) Which diagram shows complementary base pairing?

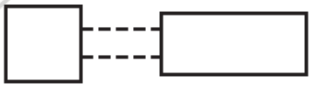
(1)

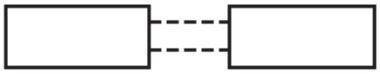
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☒ A 

☒ B 

☒ C 

☒ D 

(b) The diagram shows a sequence of bases in a DNA template (antisense) strand.

A	T	T	G	G	C	G	A	G	A	A	C

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(i) The table shows some statements about the new complementary DNA strand and the mRNA strand synthesised using
this sequence of bases.

For each statement, put **one** cross ☒ in the appropriate box, in each row, to show the correct statement about these strands.

(3)

Statement	both the new complementary DNA strand and the mRNA strand	only the new complementary DNA strand	only the mRNA strand	neither strand
The number of guanines will be the same as in the template strand	☐	☐	☐	☐
The number of thymines will be the same as the number of adenines in the template strand	☐	☐	☐	☐
There will be no adenine present	☐	☐	☐	☐

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(ii) Name the process that synthesises a mRNA strand using the DNA template strand.

(1)

.....

(Total for question = 7 marks)

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Q14.

Acute hepatic porphyria (AHP) is a very rare genetic disorder.

A drug has been developed to treat AHP.

(a) This drug was tested in a clinical trial involving 94 patients from 18 countries.

The drug was given to 48 of the patients. The other 46 patients were a control group.

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(i) Comment on the design of this clinical trial.

(2)

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(ii) Each patient was given 2.5 mg of the drug per kg of body mass, once a month.

The drug is available as a solution with a concentration of 189 mg cm^{-3}.

Calculate the volume of drug that was given each month to a patient with a body mass of 64 kg.

(2)

Answer .....

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(iii) Nausea was experienced by 27% of the patients receiving this drug.

Calculate the number of patients who experienced nausea.

(2)

Answer .....

(b) This drug is a double-stranded RNA molecule.

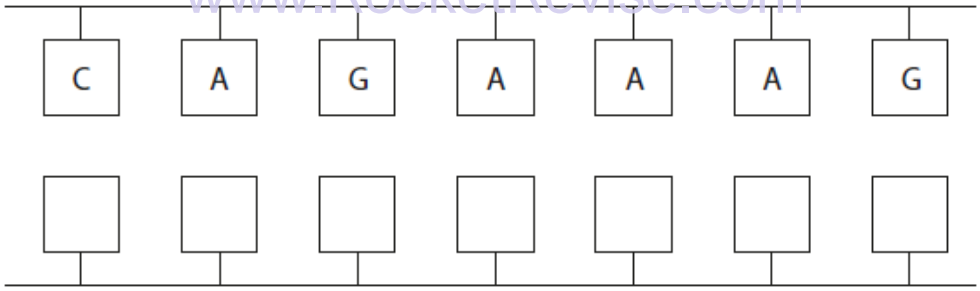
(i) The diagram shows part of the base sequence on one of the RNA strands.

Complete the diagram to show the base sequence on the other RNA strand.

(2)

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(ii) Describe the bonding in this double-stranded RNA molecule.

(3)

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(iii) In AHP, toxic porphyrin molecules build up.

The synthesis of the haem component of haemoglobin involves several steps.

Each step in the synthesis of haem is catalysed by a different enzyme.

This drug works by interfering with the mRNA copies from the gene coding for one of these enzymes.

Explain how the action of this drug helps patients with AHP.

(3)

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(Total for question = 14 marks)

Q15.

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Answer the question with a cross in the box you think is correct ☒ . If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒ .

People with lactose intolerance cannot digest lactose.

Lactose intolerance is due to a lack of the enzyme lactase.

(a) (i) Which are the products of lactose digestion?

(1)

☐ **A** fructose and galactose

☐ **B** fructose and glucose

☐ **C** galactose and glucose

☐ **D** glucose and glucose

(ii) Explain how the three-dimensional structure of lactase affects the mechanism of action of this enzyme.

(3)

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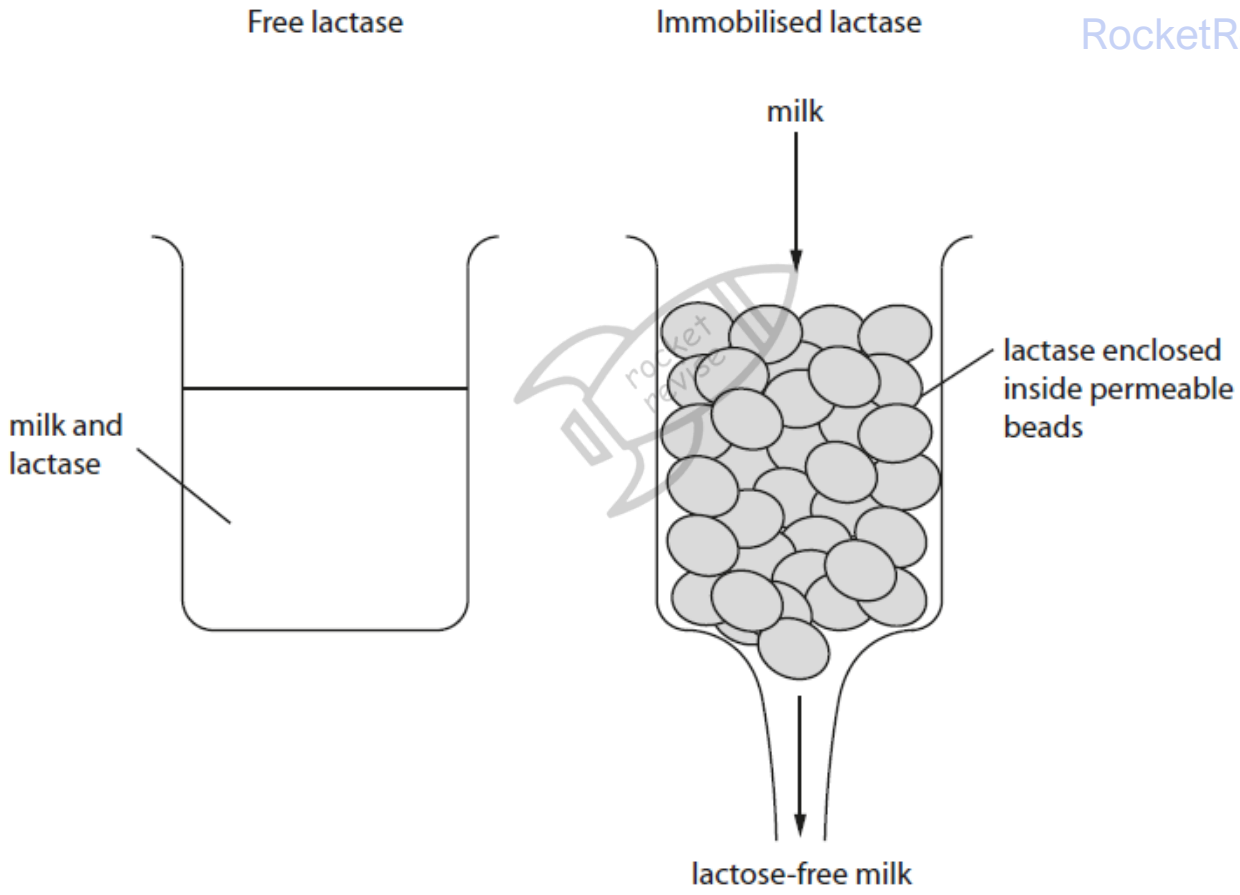
(b) People with lactose intolerance can drink lactose-free milk.

Lactose-free milk is produced by treating milk with lactase.

There are two ways of removing lactose from milk:

- mixing a solution of lactase with the milk (free lactase)
- enclosing the lactase inside permeable beads and pouring the milk over them (immobilised lactase).

The diagrams show these two methods.



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(i) Suggest **one** advantage of using immobilised lactase.

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(1)

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(ii) The table shows the effect of pH on the activity of free lactase and immobilised lactase.

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pH	Activity of free lactase / a.u.	Activity of immobilised lactase / a.u.
2	0	0
3	0	38
4	75	75
5	94	98
6	63	76
7	56	63
8	28	35

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Explain the effects of pH on the activity of these two enzymes.

(4)

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(iii) Suggest how the rate of activity of the lactase could be measured.

Include appropriate units in your answer.

(2)

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(c) Congenital lactose intolerance (CLI) is an extremely rare genetic disorder.

Most people with CLI are found in one country.

Suggest why most people with CLI are found in one country.

(2)

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(Total for question = 13 marks)

Q16.

An average-sized human chromosome contains a DNA molecule with about 300 million nucleotides, arranged in pairs.

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(a) Describe the structure of a nucleotide pair.

(3)

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(b) Before a cell can divide, the DNA molecule has to replicate. This takes place in the S phase of the cell cycle.

Human DNA replicates at a rate of 50 nucleotides per second.

Calculate how long it would take, to the nearest hour, for 150 million nucleotides to replicate.

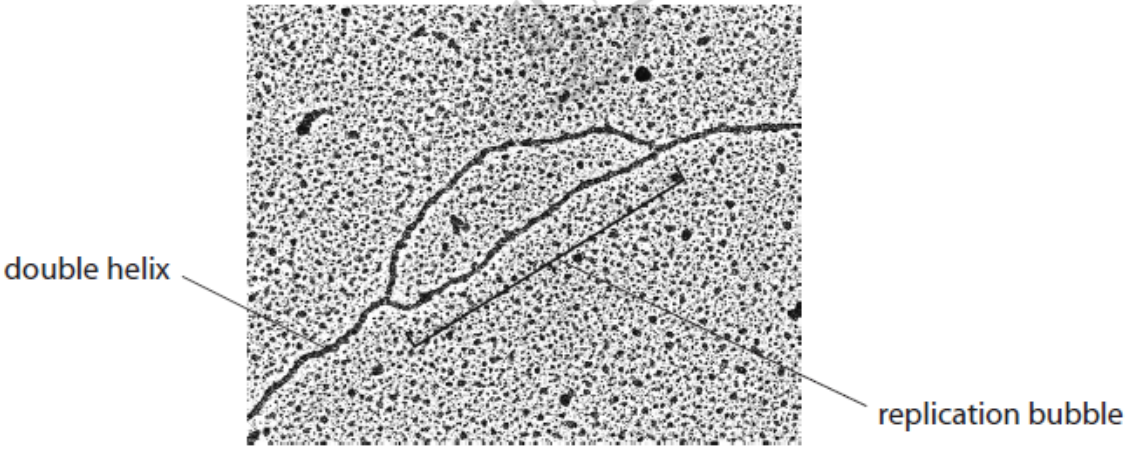
Assume that replication starts at one end of the molecule and continues to the other end.

(2)

Answer ..... hours

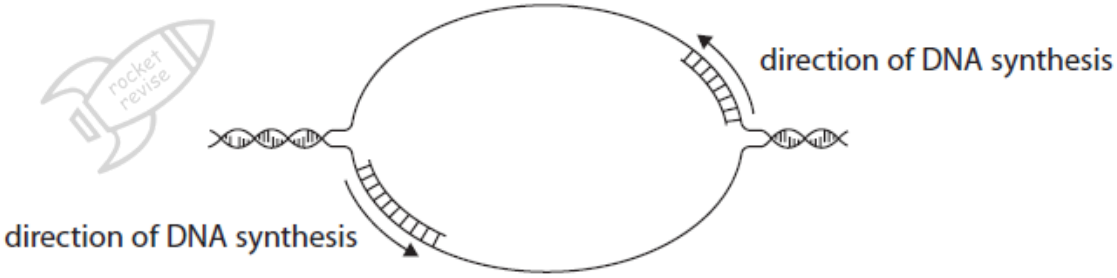
(c) In human cells, replication of DNA occurs at several sites along the molecule. These sites are called replication bubbles.

The photograph shows a replication bubble.



(Source: © Dr Gopal Murti / Science Photo Library)

The diagram shows how the DNA is replicated in one replication bubble.



(i) Explain the role of DNA polymerase in a replication bubble.

Use the information in the diagram and your own knowledge to support your answer.

(2)

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(ii) In human cells, S phase lasts about 10 hours.

Suggest why each DNA molecule is replicated using many replication bubbles.

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(3)

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(Total for question = 10 marks)

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Q17.

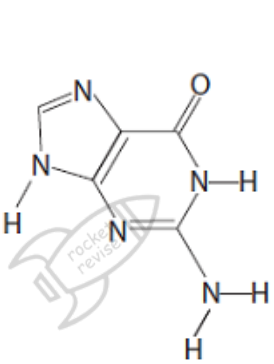
Mononucleotides are joined together to form DNA. Each mononucleotide is composed of a deoxyribose linked to a phosphate and one of four bases.

(a) The diagrams show the structure of the four bases.

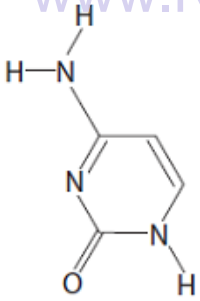
Guanine and adenine have two ring structures, cytosine and thymine have one ring structure.

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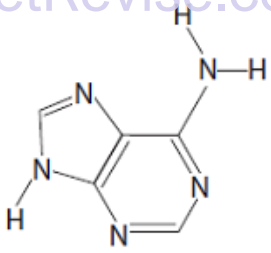
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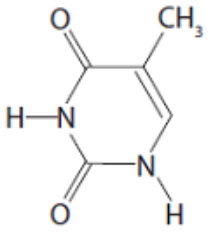
guanine



cytosine



adenine



thymine

Explain the role of these bases in the structure of DNA.

Use the information in the diagram to support your answer.

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(3)

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(b) The DNA of the bacterium *E. coli* has a mass of 3.1×10^9 daltons (Da) and consists of 4.7×10^6 base pairs.

(i) Calculate the mean mass of a base pair.

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Express your answer in standard form.

(1)

Answer ..... Da

(ii) Calculate the total mass of DNA produced after three cell divisions.

Give your answer in grams, where 1 dalton = 1.67×10^{-24} grams.

(2)

Answer ..... g

(c) Meselson and Stahl provided data that supported the theory that DNA replicates by

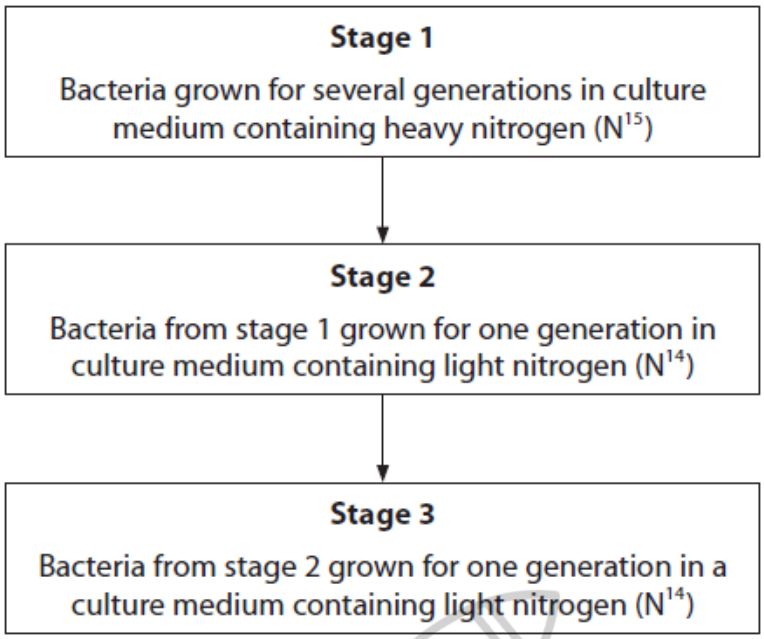
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semi-conservative replication and disproved competing theories.

Heavy nitrogen (N^{15}) and light nitrogen (N^{14}) were used in their experiments.

The flow chart summarises part of one experiment performed by Meselson and Stahl.



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After each stage, a sample of DNA was taken from the bacteria and the DNA molecules separated on a density gradient in a tube.

The heavier DNA molecules form bands lower down the gradient than the lighter DNA molecules.

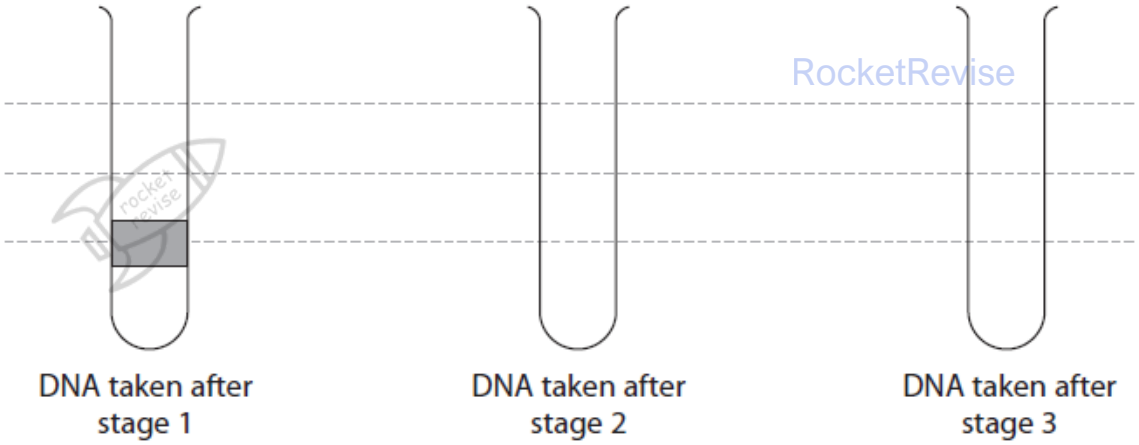
The width of each band is proportional to the percentage of molecules in the sample.

(i) Complete the diagram to show the results of this investigation.

Use the dotted lines to help you position the bands on the diagram.

The first one has been done for you.

(3)



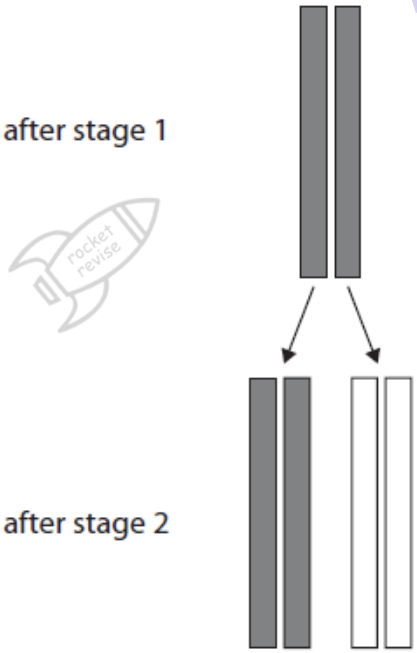
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(ii) One competing theory was the conservative theory.

The diagram shows the conservative theory for the replication of DNA.

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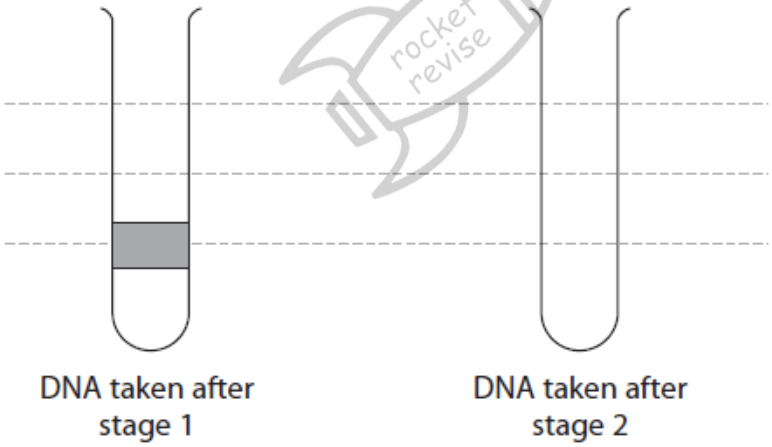
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Complete the diagram to show the position of the bands on the density gradient in a tube if this was the correct theory.

(2)



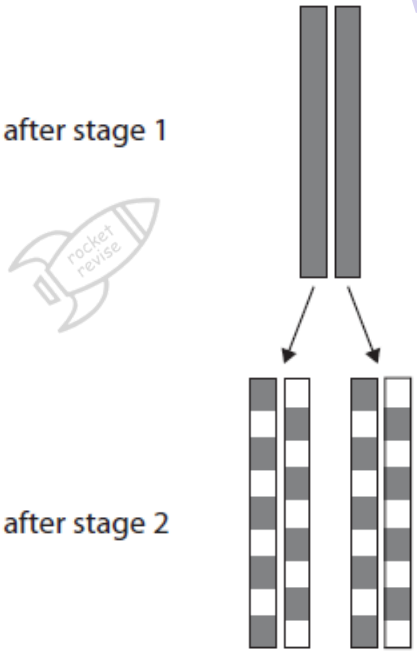
(iii) Another competing theory was the dispersive theory.

The diagram shows the dispersive theory.

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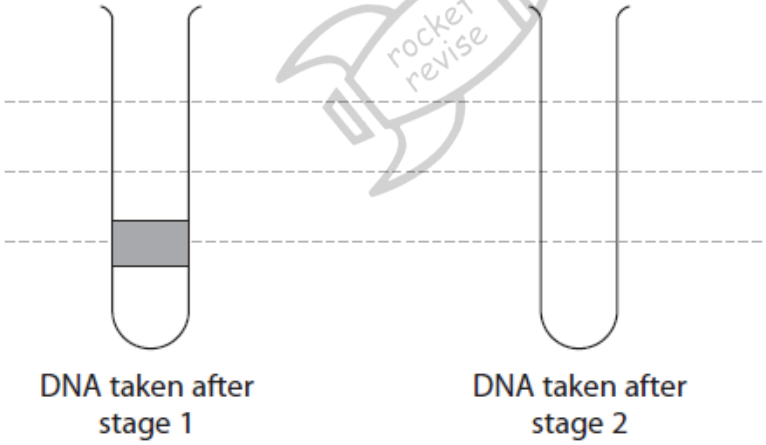
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Complete the diagram to show the position of the bands on the density gradient in a tube if this was the correct theory.

(2)



(Total for question = 13 marks)

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Q18.

Answer the questions with a cross in the boxes you think are correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Pectinase is an enzyme that breaks down pectin. Pectin is present in the cell walls of fruits.

(a) What type of molecule is pectinase?

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(1)

☒ **A** disaccharide

☒ **B** polysaccharide

☒ **C** protein

☒ **D** triglyceride

(b) Explain why enzymes are described as biological catalysts.

(2)

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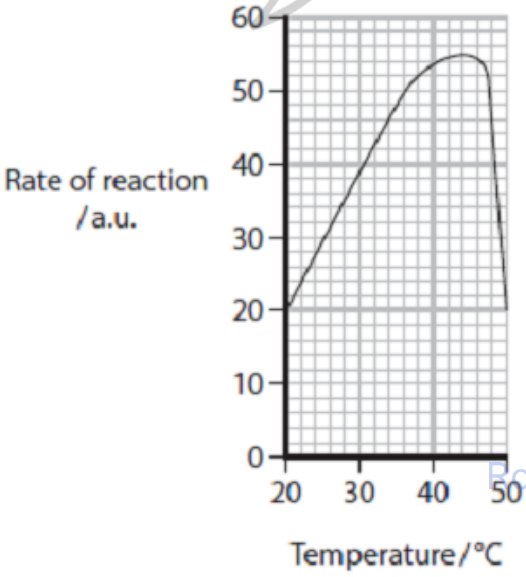
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(c) The graph shows the effect of temperature on the rate of reaction of pectinase.



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(i) Calculate the effect on the rate of reaction of each degree increase in temperature between 46°C and 50°C.

Give your answer to two significant figures.

(1)

Answer ..... a.u. per °C

(ii) State why there is a decrease in the rate of reaction above the optimum temperature.

(1)

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(Total for question = 5 marks)

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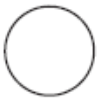
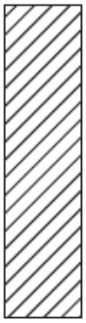
Q19.

Answer the questions with a cross in the boxes you think are correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

The components of a cell membrane determine its properties.

The membrane is a phospholipid bilayer.

(a) The table shows one way that the components of phospholipids can be drawn.

Component	Way of drawing the component
phosphate group	
glycerol	
fatty acid	
bond	

(i) Draw a diagram of a phospholipid, using the information in the table.

(2)

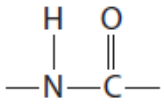
(ii) One of the bonds present in phospholipids is an ester bond.

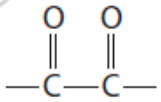
Which diagram shows an ester bond?

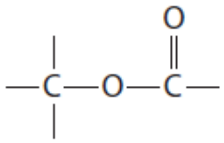
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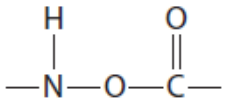
(1)

☒ **A** 

☒ **B** 

☒ **C** 

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☒ **D** 

(b) The fluidity of a membrane is increased by a number of factors:

- increase in temperature
- increase in the proportion of unsaturated fatty acids
- decrease in cholesterol
- fatty acids with shorter side chains.

(i) Explain why an increase in temperature increases the fluidity of the membrane.

(2)

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(ii) Which row in the table describes an unsaturated fatty acid compared with a saturated fatty acid with the same number of carbon atoms?

(1)

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	Has double bonds between two carbons	Number of hydrogens
☒ A	no	fewer than the saturated fatty acid
☒ B	no	more than the saturated fatty acid
☒ C	yes	fewer than the saturated fatty acid
☒ D	yes	more than the saturated fatty acid

(iii) Explain why a decrease in cholesterol increases the fluidity of the membrane.

(2)

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(iv) Suggest why a fatty acid with a shorter side chain will increase the fluidity of a membrane.

(1)

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(c) Studies have shown that when fish are moved into water at a lower temperature, the fatty acid content of their membranes changes.

(i) Suggest how the content of the fatty acids in these membranes may have changed.

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(2)

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(ii) Suggest why these changes are necessary for the survival of the fish.

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(2)

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(Total for question = 13 marks)

Q20.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Enzymes are biological catalysts that have an effect on the energy required for a reaction.

(a) State the meaning of the term **biological catalyst**.

(2)

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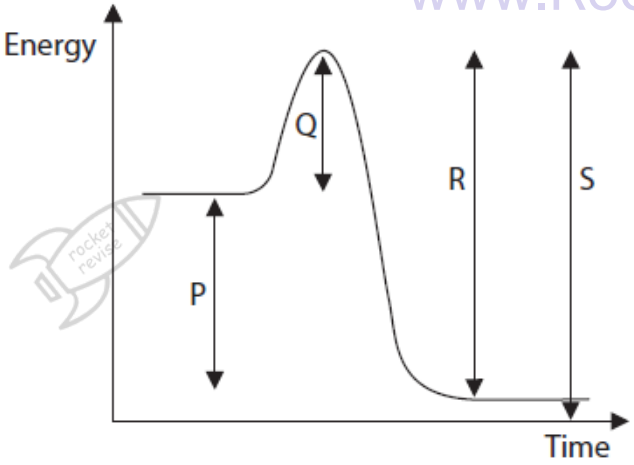
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(b) The graph shows the energy changes during an enzyme catalysed reaction.

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Which arrow shows the energy required to start the reaction?

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(1)

☒ **A** P

☒ **B** Q

☒ **C** R

☒ **D** S

(c) The diagram shows a series of chemical reactions.

$$A + B \xrightarrow{E_1} AB \xrightarrow{E_2} AC \xrightarrow{E_3} D$$

Three different enzymes, E_1, E_2 and E_3, are involved in these reactions.

Explain why E_1 cannot convert AB into AC or AC into D.

Use the information in the diagram to support your answer.

(3)

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(d) Pectinase is an enzyme that breaks down pectin.

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The effect of different combinations of pH and temperature on pectinase activity was investigated.

The volume and concentration of pectinase and pectin were kept the same in all reactions.

The table shows the rate of reaction for each pH and temperature combination tested.

pH	Temperature / °C	Rate of reaction / a.u.
2	5	0
2	30	0
2	60	0
4	5	5
4	30	35
4	60	0
8	5	5
8	30	10
8	60	0

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Describe **two** conclusions that can be drawn from these results.

(2)

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(Total for question = 8 marks)

Q21.

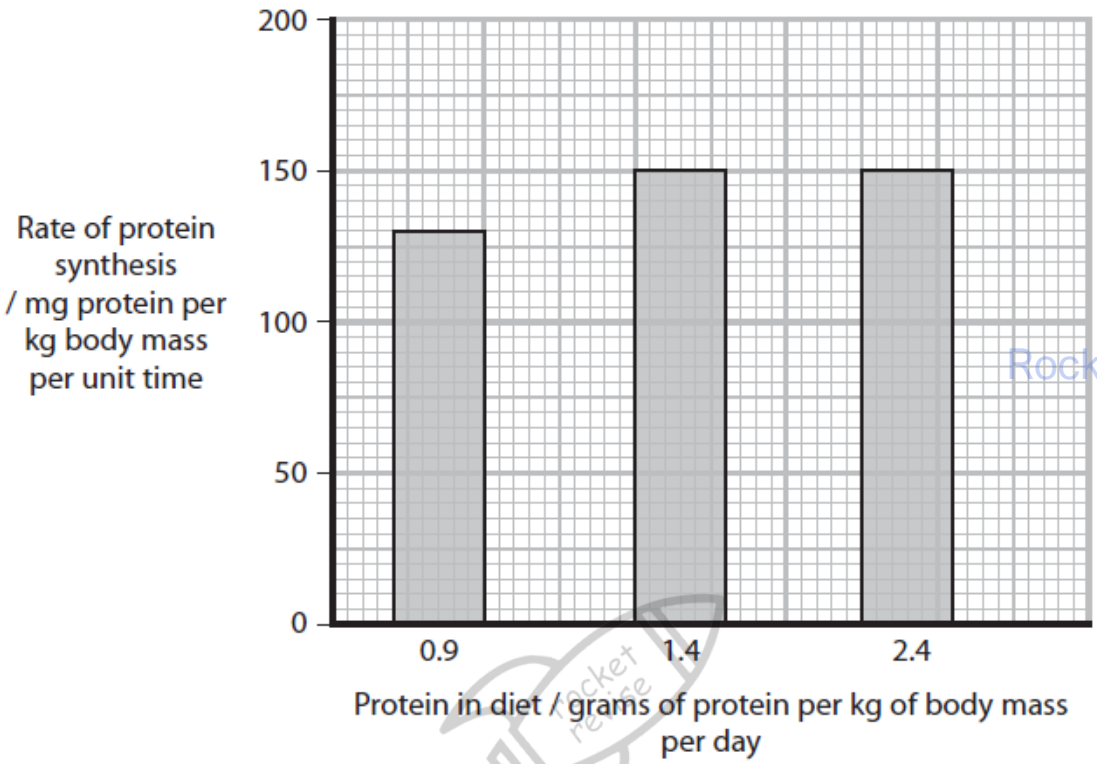
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Amino acids in the diet of animals are used to synthesise proteins.

(a) The rate of protein synthesis is dependent on a number of factors.

The graph shows the effect of the mass of protein in the diet on the rate of protein synthesis.



Calculate the minimum mass of protein a person with a mass of 70 kg should eat in one week if they want a rate of protein synthesis of 150 mg protein per kg body mass per unit time.

(2)

Answer ..... g

(b) (i) Name the **two** stages in protein synthesis, in the order that they occur.

(1)

Stage 1

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Stage 2

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(ii) Describe the roles of messenger RNA (mRNA) and transfer RNA (tRNA) in protein synthesis.

(4)

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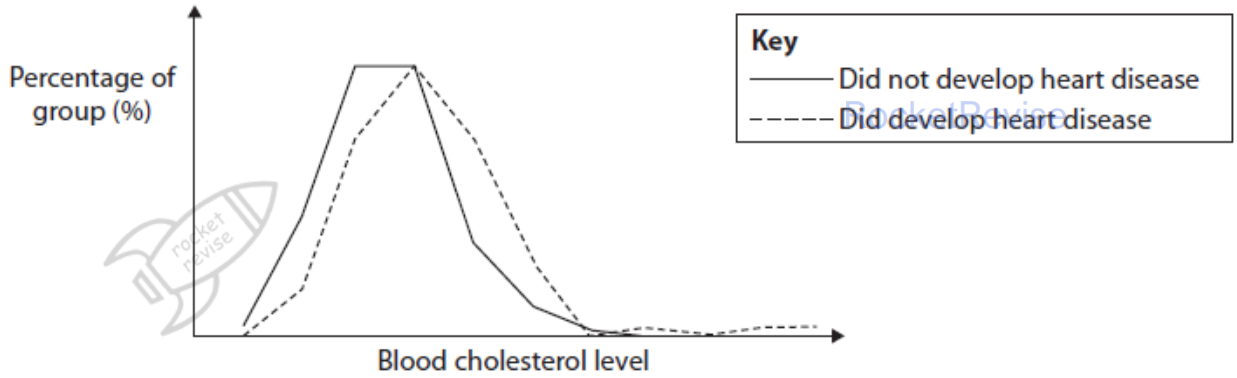
(Total for question = 7 marks)

Q22.

A number of factors increase the risk of cardiovascular disease (CVD), including blood cholesterol levels.

(a) In one investigation, scientists measured the blood cholesterol level in a group of people and monitored the development of heart disease in the following 10 years.

The graph shows the results of this investigation.



Some people interpreted the data as showing evidence for a relationship between blood cholesterol levels and heart disease.

Other people interpreted the data as not showing evidence for a relationship between blood cholesterol levels and heart disease.

Describe the evidence for and against a relationship between blood cholesterol levels and heart disease.

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(2)

Evidence for

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Evidence against

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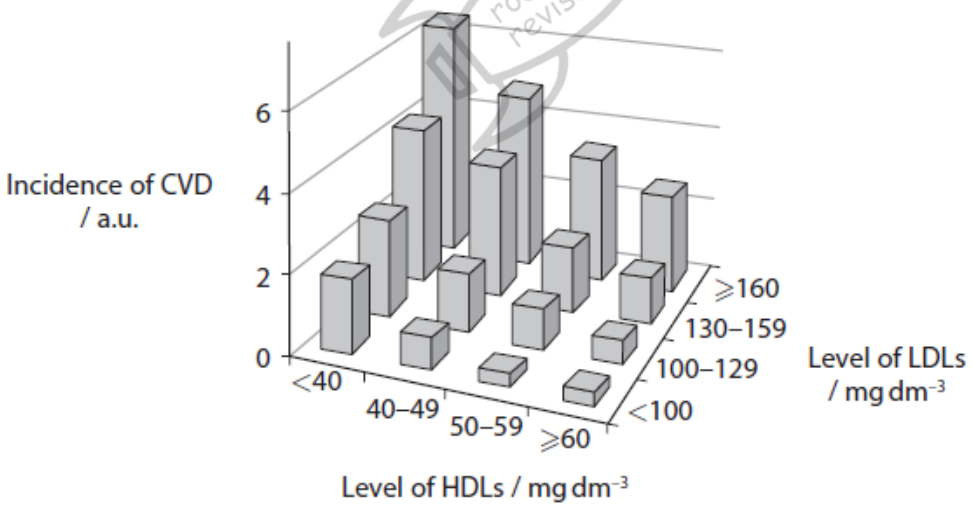
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(b) Other investigations have shown that the levels of high-density lipoproteins (HDLs) and low-density lipoproteins (LDLs) are significant in determining the risk of CVD.

The graph shows the relationship between levels of HDLs and LDLs on the incidence of CVD.



The following conclusions can be made from this graph:

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- as the levels of LDLs increases so does the incidence of CVD
- as the levels of HDLs increases the incidence of CVD decreases
- the higher the ratio of HDLs to LDLs the lower the incidence of CVD.

Describe the evidence shown in this graph that support these conclusions.

(3)

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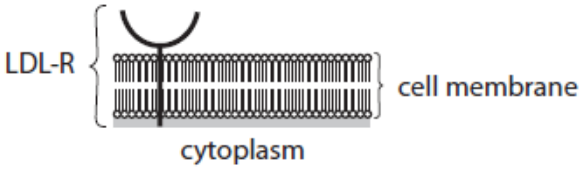
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(c) The low-density lipoprotein receptor (LDL-R) is a protein on the surface of liver cells that is involved in the endocytosis of LDLs.

The diagram shows an LDL-R in a cell membrane.

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(i) Explain how the LDL-R is involved in endocytosis of LDLs.

(3)

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(ii) The lower pH inside the cell causes the LDL to separate from its receptor.

Suggest how a change in pH could cause this separation.

(2)

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(iii) Mutations in the gene coding for the LDL-R can result in an increased risk of CVD.

Explain why mutations in this gene can increase the risk of CVD.

(3)

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(Total for question = 13 marks)

Q23.

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Nucleic acids include DNA and RNA.

(a) Read through the following passage about DNA.

Complete the passage by writing the most appropriate word or words on the dotted lines.

(5)

A strand of DNA is composed of mononucleotides linked by ..... bonds formed during ..... reactions.

During this type of reaction a molecule of ..... is also formed.

One DNA molecule has two strands of DNA joined together by

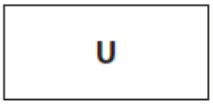
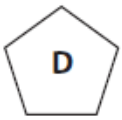
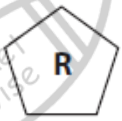
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..... bonds between complementary base pairs.

The DNA molecule then twists to form a ..... .

(b) The table shows some components of DNA mononucleotides and RNA mononucleotides and shapes that can be used to represent them.

Component	Shapes
bond	
thymine	
uracil	
deoxyribose	
ribose	
phosphate	

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Draw **one** DNA mononucleotide, using shapes selected from those shown in the table.

(3)

(Total for question = 8 marks)

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Q24.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Polyphenol oxidase is an enzyme found in avocados.

This enzyme causes green avocados to go brown when they are cut open.

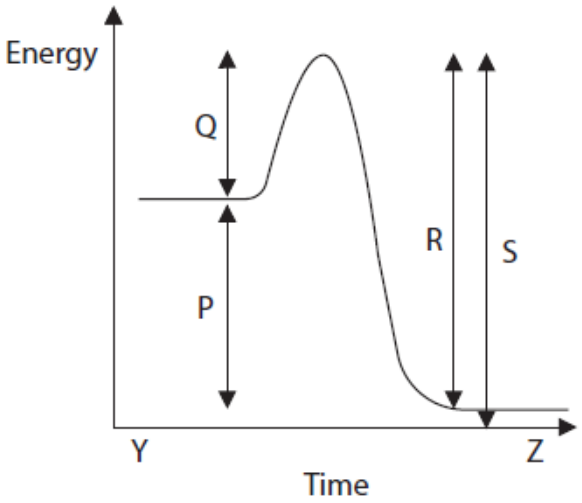
The equation shows the reaction catalysed by this enzyme.

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monophenol ⟶ *o*-diphenol

(a) The graph shows the energy changes during this reaction, when the enzyme is not present.



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(i) Which arrow shows the activation energy for this reaction?

(1)

☒ **A** P

☒ **B** Q

☒ **C** R

☒ **D** S

(ii) Which row of the table shows the molecule present at time Y and time Z on this graph?

(1)

	Time Y		Time Z	
☒ **A**	product	monophenol	reactant	*o*-diphenol
☒ **B**	product	*o*-diphenol	reactant	monophenol
☒ **C**	reactant	monophenol	product	*o*-diphenol
☒ **D**	reactant	*o*-diphenol	product	monophenol

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(iii) Which row of the table shows how the length of the arrows P and Q on this graph would change when polyphenol oxidase is present?

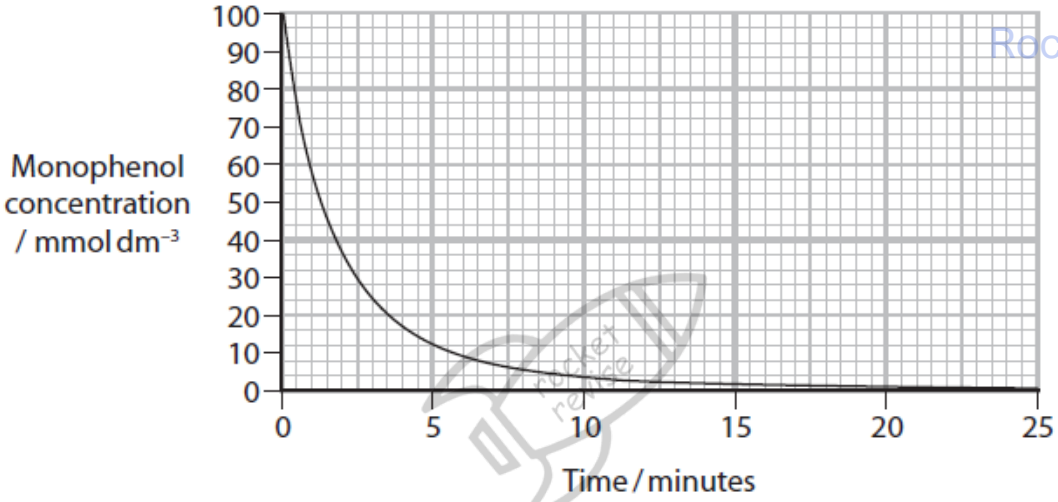
(1)

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		Change in length of arrow	
		P	Q
☒	A	increase	decrease
☒	B	increase	no change
☒	C	no change	decrease
☒	D	no change	no change

(b) The graph shows the change in monophenol concentration in the presence of polyphenol oxidase, at a temperature of 25 °C.



(i) Calculate the rate of reaction at 3 minutes.

Draw a tangent to help with your calculation.

(2)

Answer ..... mmol $dm^{-3}min^{-1}$

(ii) This reaction was carried out below the optimum temperature for the enzyme.

Explain what would happen to the shape of this curve if the temperature was increased to the optimum temperature for this enzyme.

(2)

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(Total for question = 7 marks)

Q25.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Messenger RNA (mRNA) and transfer RNA (tRNA) are involved in protein synthesis.

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(a) The diagram shows the sequence of bases in part of a DNA template (antisense) strand and the mRNA molecule formed during protein synthesis.

DNA template strand	A	T	G	C	C	G	T	T
mRNA								

(i) Complete the diagram to show the sequence of bases in this mRNA.

(1)

(ii) Name this stage of protein synthesis.

(1)

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(iii) Name the enzyme used in this stage of protein synthesis.

(1)

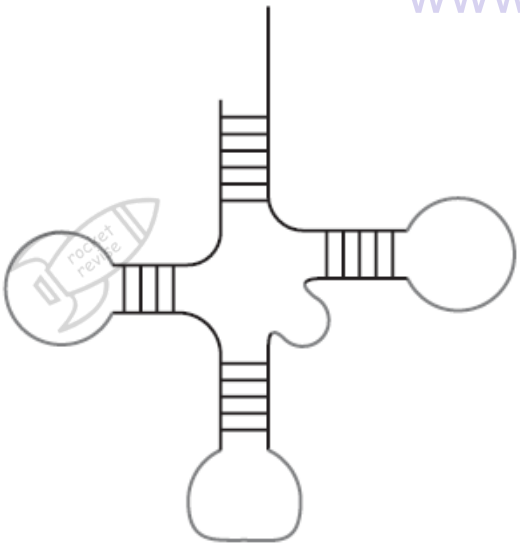
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(b) The diagram shows part of a tRNA molecule.

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Compare and contrast the structure of an mRNA molecule with that of a tRNA molecule.

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(3)

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(Total for question = 6 marks)

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Q26.

Elastin and collagen are proteins found in connective tissue.

(a) Describe the structure of collagen.

(3)

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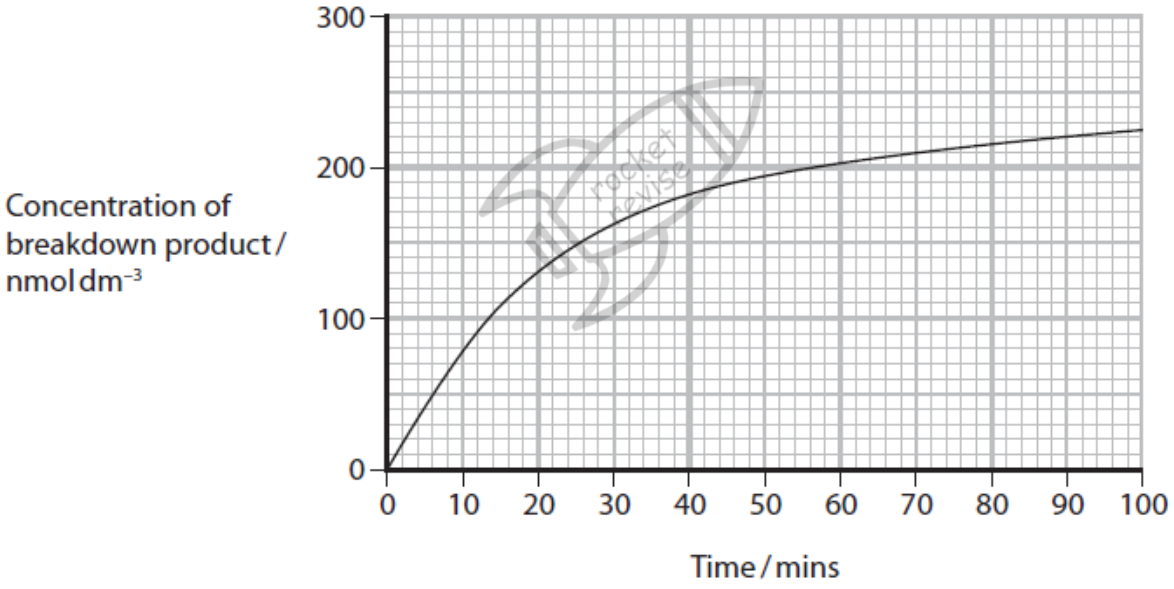
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(b) Proteases are enzymes that break down proteins.

The graph shows the effect of a protease solution on a sample of elastin.



(i) Calculate the rate of reaction at 30 minutes.

Draw a tangent to help with your calculation.

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(2)

Answer ..... nmol dm^{-3} min^{-1}

(ii) Explain how a protease breaks down elastin.

(3)

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(Total for question = 8 marks)

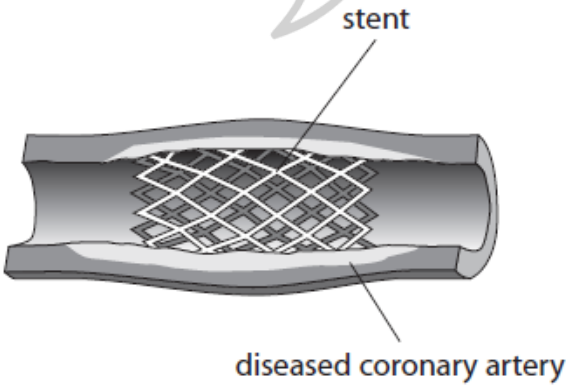
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Q27.

Stents are used in the treatment of atherosclerosis.

Stents widen the diseased coronary artery so that blood can flow through to the heart muscle.

The diagram shows a stent.



Stents can damage the endothelial cells lining the artery and trigger the formation of a blood clot.

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Patients who have a stent fitted are given anticoagulants.

(a) Explain how damage to the endothelial cells results in the formation of thrombin.

(2)

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(b) Give **one** risk of treating patients with anticoagulants.

(1)

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(c) The photograph shows a tropical bont tick.

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(Source: © Phanie/Alamy Stock Photo)

This bont tick produces a peptide called variegin.

Variegin binds to the active site of thrombin and prevents blood clotting.

(i) Variegin is a peptide consisting of 32 amino acids.

Which is the minimum number of bases in the gene that codes for this peptide?

(1)

☒ **A** 32

☒ **B** 33

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☒ **C** 96

☒ **D** 99

(ii) Explain how variegin prevents blood clotting.

(2)

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(iii) Thrombin slowly splits variegin into two molecules.

Explain how thrombin splits variegin.

(2)

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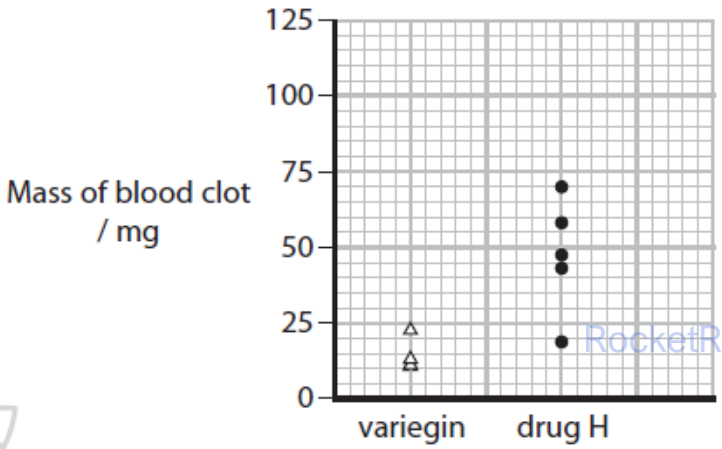
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(iv) The effect of variegin on the mass of blood clots forming inside a stent was investigated.

The masses were compared with another anticoagulant, drug H.

The graph shows the results of this investigation.

Each plotted point represents the mass of blood clot in one stent.



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Comment on the results of this investigation.

(2)

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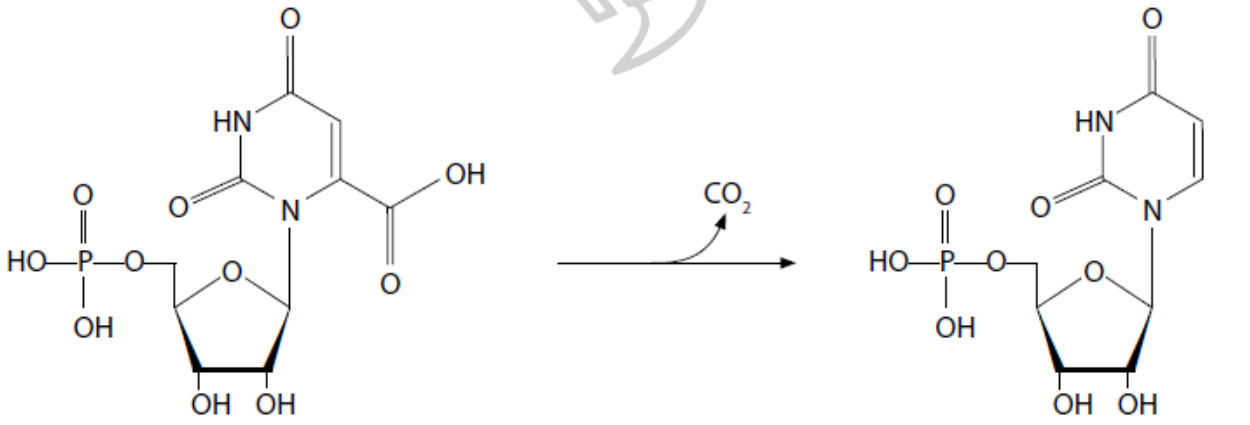
(Total for question = 10 marks)

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Q28.

The enzyme OMP decarboxylase is involved in the synthesis of the mononucleotide uridine monophosphate.

(a) The diagram shows the reaction catalysed by OMP decarboxylase.



orotidine monophosphate (OMP) uridine monophosphate

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(i) Which sugar is present in uridine monophosphate?

(1)

- ☒ **A** deoxyribose
- ☒ **B** galactose
- ☒ **C** glucose
- ☒ **D** ribose

(ii) Draw a circle around the base in uridine monophosphate.

(1)

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(iii) Suggest one way in which cells use uridine monophosphate.

(1)

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(b) The enzyme OMP decarboxylase increases the rate of carbon dioxide removal from orotidine monophosphate by 10^{17} times.

(i) State how OMP decarboxylase increases the rate of this reaction.

(2)

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(ii) Explain why OMP decarboxylase catalyses this reaction only.

(3)

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(Total for question = 8 marks)

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Q29.

Collagen is an insoluble, fibrous protein.

(a) Describe the roles of RNA in the synthesis of collagen.

(5)

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(b) Explain how the primary structure of collagen determines its properties.

(4)

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(Total for question = 9 marks)

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Q30.

Red-green colour blindness is a common trait in humans.

(a) The gene for red-green colour blindness is located on the X chromosome.

State what is meant by the term **gene**.

(1)

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(b) Describe how the two strands of DNA forming the double helix in a gene are held together.

(2)

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(c) Explain why each codon for the DNA genetic code must contain at least three bases.

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(d) A red-green colour blind father and an unaffected heterozygous mother had a child.

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Determine the probability of this child being red-green colour blind.

Use a genetic diagram to support your answer.

(3)

Probability .....

(Total for question = 9 marks)

Q31.

Glucose is an example of a monosaccharide.

Monosaccharides join together to form disaccharides, oligosaccharides and polysaccharides.

Oligosaccharides contain from three to ten monosaccharides.

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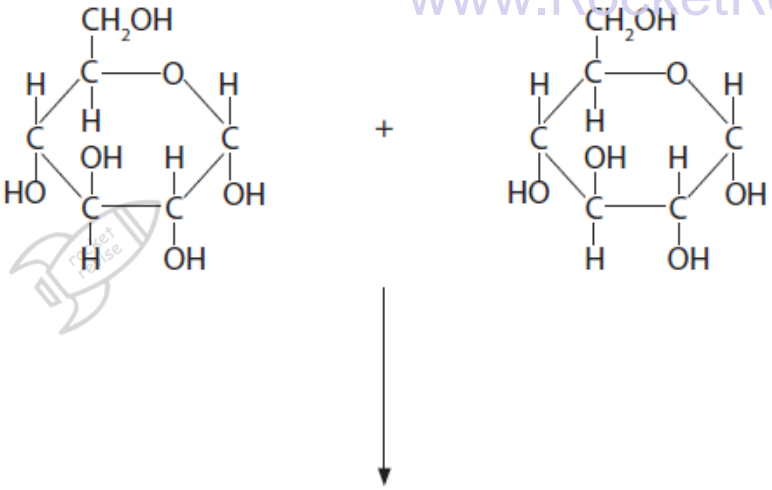
(a) The diagram shows two glucose molecules.

Complete the diagram to show how these two molecules join by a glycosidic bond to form **two** products.

(3)

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(b) Starch is a polysaccharide composed of amylose and amylopectin.

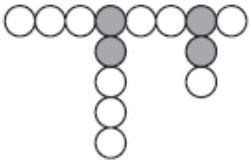
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Digestion of starch begins in the mouth by an enzyme called amylase.

This amylase can break only 1–4 glycosidic bonds.

Only disaccharides and oligosaccharides are produced during this reaction.

(i) The diagram shows part of an amylopectin molecule.



Key

○ glucose molecule joined by 1–4 glycosidic bonds

● glucose molecule joined by 1–4 and 1–6 glycosidic bonds

Draw **one** disaccharide and **two** different oligosaccharides that could be produced following the digestion of this part of amylopectin by this amylase.

(2)

(ii) Explain why this amylase is able to break only 1–4 glycosidic bonds.

(2)

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(c) The digestion of carbohydrates is completed in the small intestine.

Glucose is then absorbed into the bloodstream.

(i) Suggest how disaccharides and oligosaccharides are broken down in the small intestine.

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Give reasons for your answer.

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(ii) The absorption of glucose into the bloodstream requires membrane transport proteins.

Explain why glucose cannot diffuse into the bloodstream.

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(Total for question = 12 marks)

Q32.

Water is an important molecule in living organisms.

(a) Water is a dipolar molecule that forms hydrogen bonds with other water molecules.

(i) Explain the dipole nature of water.

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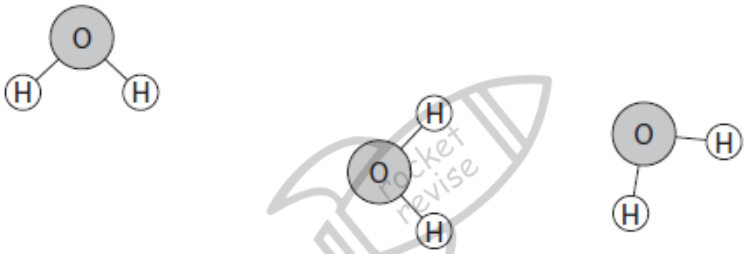
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(ii) The diagram shows three water molecules.

Complete the diagram to show a hydrogen bond between two water molecules.

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(1)



(b) Water plays a role as a solvent.

Sodium chloride dissolves in water.

The diagram shows a sodium ion (Na^+).

Complete the diagram to show the arrangement of two water molecules around a sodium ion when it is dissolved.

(1)



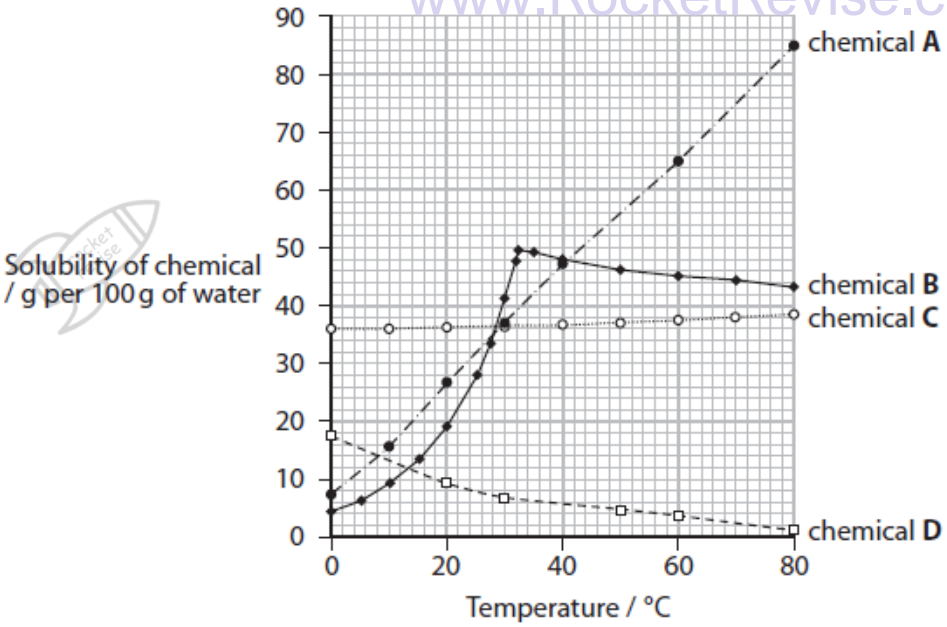
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(c) Temperature affects the solubility of chemicals in water.

The graph shows the effect of temperature on the solubility of four chemicals, **A**, **B**, **C** and **D**.

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Describe the effects of temperature on the solubility of these chemicals.

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(Total for question = 8 marks)

Q33.

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The cell membrane consists of phospholipids and proteins.

(a) Name the current model for the structure of the cell membrane.

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(b) Describe the arrangement of phospholipids in the cell membrane in this model.

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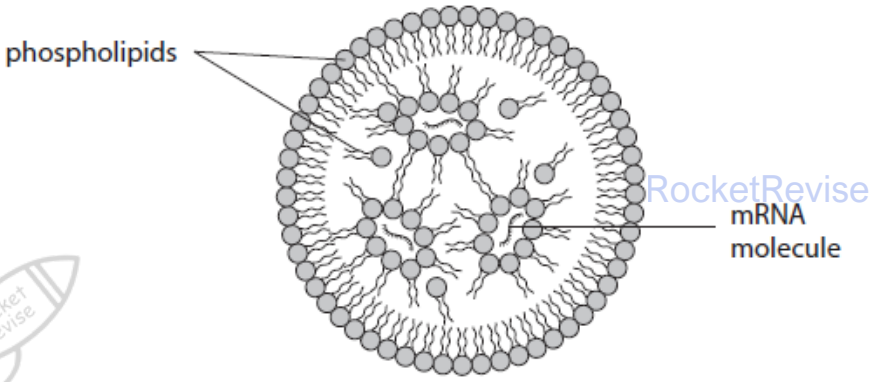
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(c) Phospholipids can be used to form lipid nanoparticles.

Lipid nanoparticles can be used to carry nucleic acids to tissues in the body.

The diagram shows a lipid nanoparticle containing messenger RNA (mRNA) molecules.



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(i) Lipid nanoparticles are very small.

Calculate the diameter of a lipid nanoparticle with a volume of 0.108 μm^3.

Use the formula

$$V = \frac{4}{3}\pi r^3$$

Use π = 3.0

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(3)

Answer ..... µm

(ii) Describe the structure of a mRNA molecule.

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(iii) Explain the arrangements of phospholipids in this lipid nanoparticle that carries mRNA to tissues in the body.

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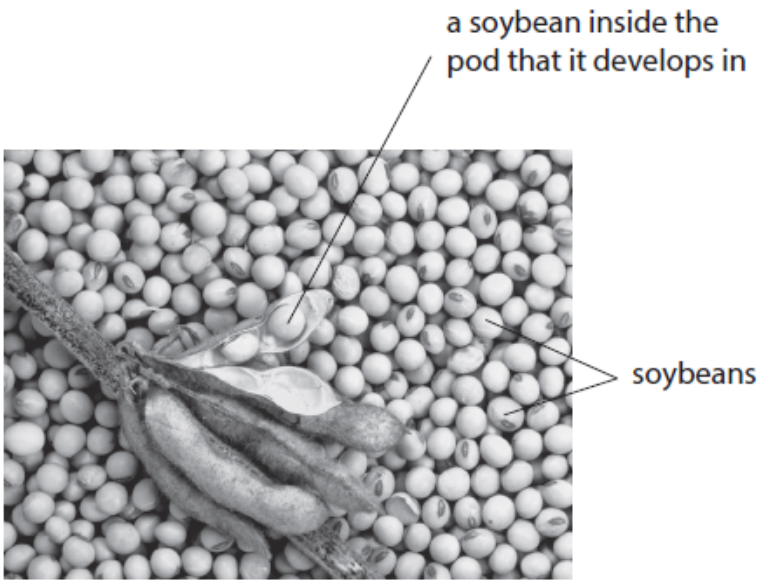
(Total for question = 13 marks)

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Q34.

The photograph shows some soybeans.



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(a) Eating soybeans may reduce cholesterol and lowâ€‘density lipoproteins (LDLs) in the blood.

Explain why eating soybeans may reduce the risk of heart disease.

(3)

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(b) The extent to which soybeans reduce cholesterol depends on the variety of soybeans. This reduction is influenced by two proteins found in soybeans, glycinin and β-conglycinin.

Give **two** ways that the primary structure of two proteins could differ from each other.

(1)

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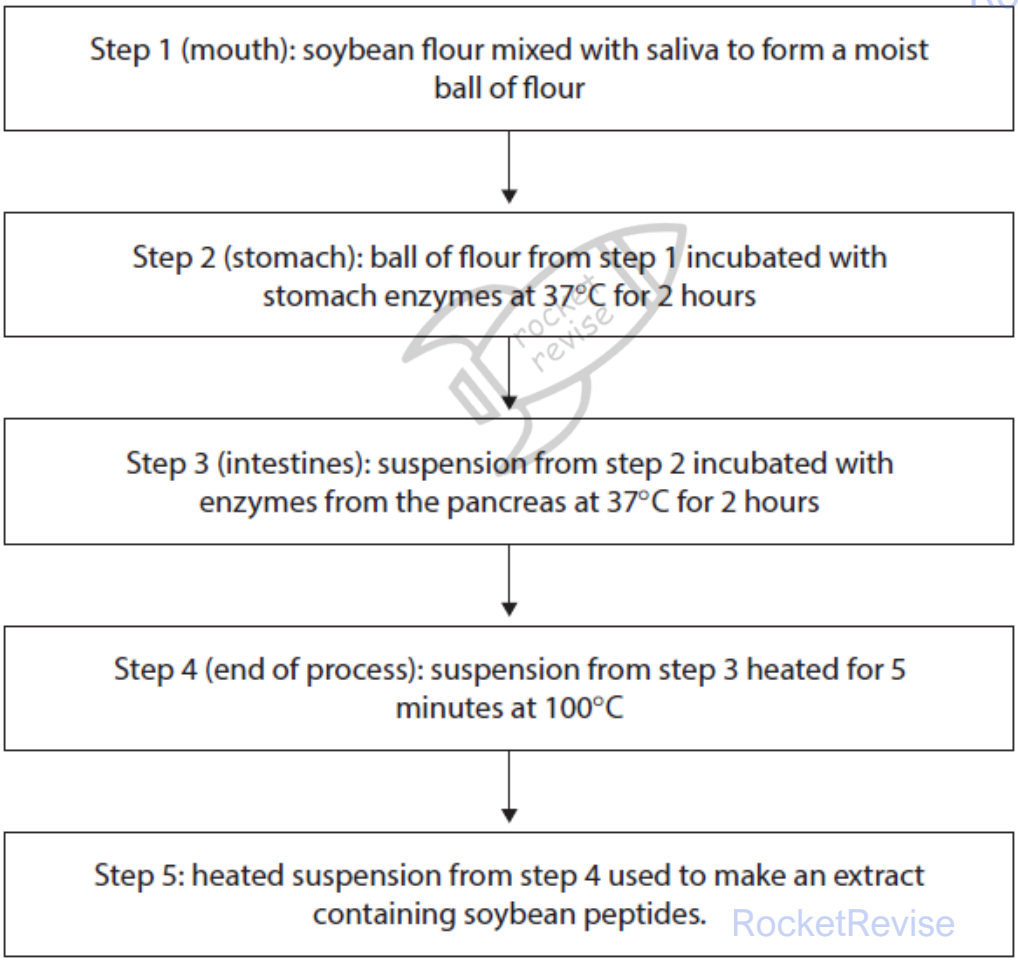
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(c) The effect of soybean extracts on cholesterol production by liver cells growing in culture was investigated.

Scientists produced fat-free soybean flour and digested it with enzymes, in a way that copied digestion in the human digestive system.

The diagram shows some of the steps used.

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(i) Explain why the scientists used digested soybeans to make the extract for this investigation.

(2)

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(ii) Suggest why the soybean flour was incubated with different enzymes in steps 2 and 3.

(1)

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(iii) Explain why the soybean flour and enzymes were incubated at 37 °C for 2 hours in steps 2 and 3.

(2)

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37 °C

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2 hours

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(d) The soybean peptides, at different concentrations, were incubated with the liver cells growing in culture.

The enzyme HMGCR is involved in the synthesis of cholesterol.

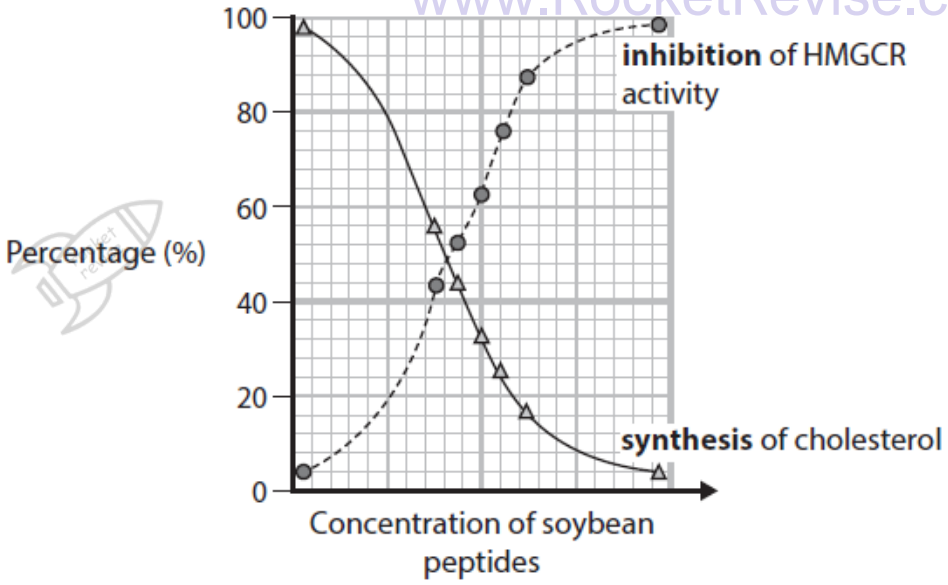
The effect of soybean peptides on the activity of this enzyme and the synthesis of cholesterol was measured.

The graph shows the results.

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Explain why eating soybeans may reduce cholesterol levels in the blood.

Use the information in the graph to support your answer.

(3)

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(Total for question = 12 marks)

Q35.

The process of protein synthesis involves RNA polymerase, start and stop codons and ribosomes.

(a) Describe the role of RNA polymerase in protein synthesis.

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(b) Compare and contrast the structure of the start and stop codons and how they work.

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(c) Describe the roles of the ribosomes in protein synthesis.

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(Total for question = 7 marks)

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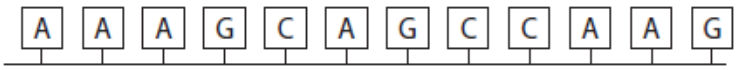
Q36.

The nature of the genetic code of a gene is important in determining the sequence of amino acids in a polypeptide chain.

The diagrams show the genetic code for some amino acids and the sequence of bases in part of a gene.

Genetic code	Amino acid	Genetic code	Amino acid	Genetic code	Amino acid	Genetic code	Amino acid
AAA AAG	Lysine	CAA CAG	Glutamine	GAA GAG	Glutamic acid	TAC TAT	Tyrosine
AAC AAT	Asparagine	CAT CAC	Histidine	GAC GAT	Aspartate	TCA TCC TCG TCT	Serine
ACA ACC ACG ACT	Threonine	CCA CCC CCG CCT	Proline	GCA GCC GCG GCT	Alanine	TGG	Tryptophan

Part of a gene:



The genetic code is described as a triplet code that is non-overlapping and is degenerate.

Explain what each of these phrases means.

Use the information provided to illustrate your answer.

(a) Triplet code

(2)

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(b) Non-overlapping code

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(2)

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(c) Degenerate code

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(2)

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(Total for question = 6 marks)

Q37.

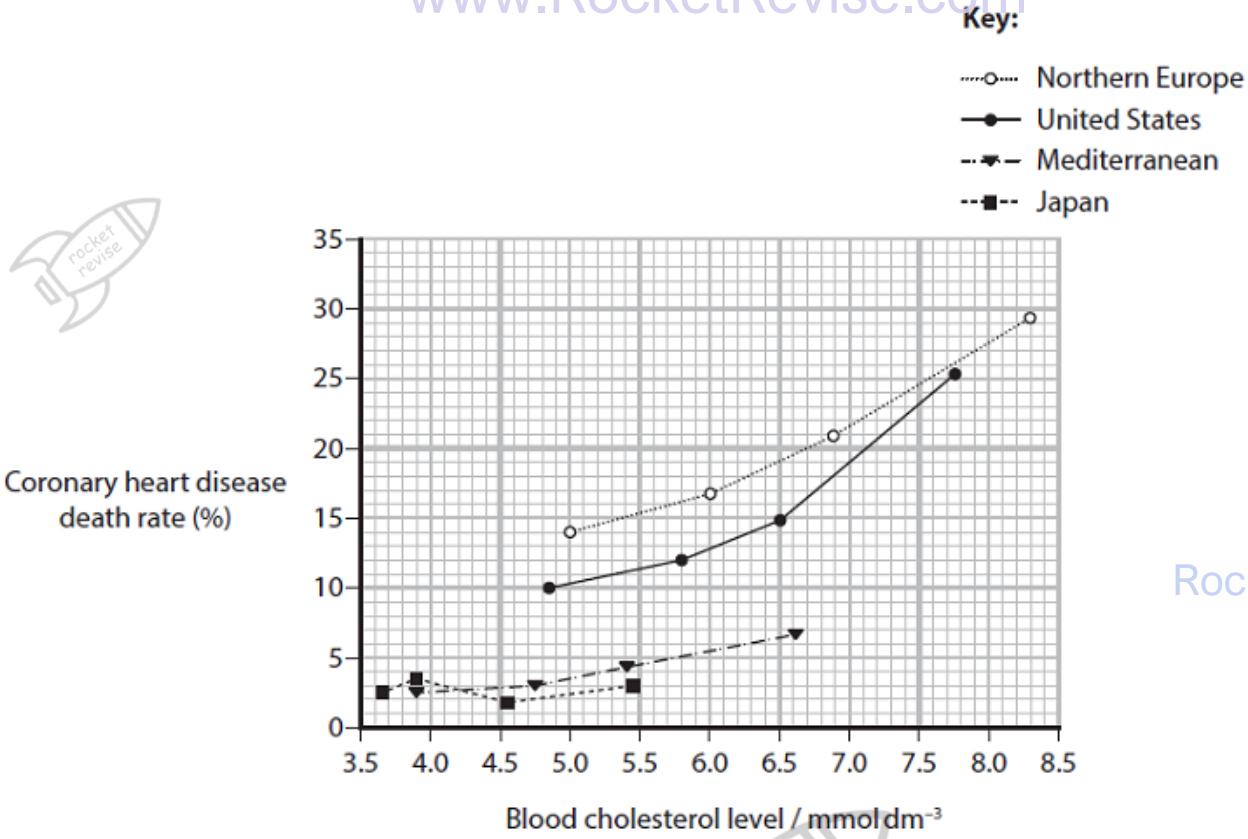
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A high blood cholesterol level is associated with an increased risk of cardiovascular disease (CVD).

(a) The graph shows the relationship between blood cholesterol levels and the coronary heart disease death rate in men in four parts of the world.

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Describe **two** conclusions that can be made about coronary heart disease death rate in these parts of the world.

(2)

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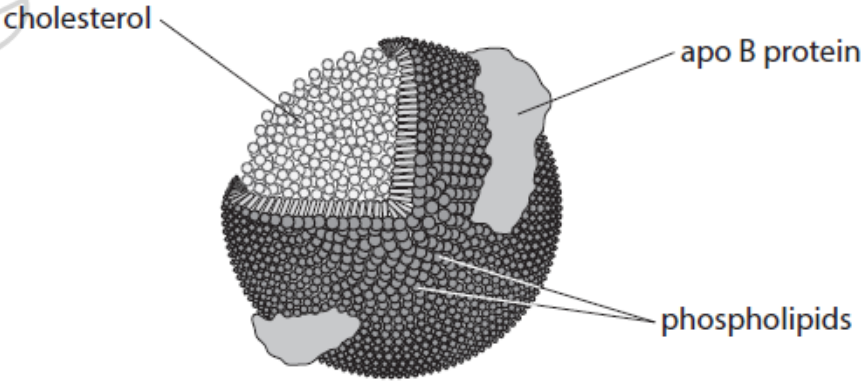
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(b) Cholesterol can be transported in the blood as low-density lipoproteins (LDLs) that are made in the liver.

The diagram shows a LDL.



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Statins can be used to reduce blood cholesterol levels.

Statins reduce cholesterol synthesis, reduce the synthesis of apo B protein and increase the number of LDL receptors on the surface of liver cells.

(i) Statins can cause side effects.

Give **two** side effects of taking statins.

(1)

1 .....

2 .....

(ii) Liver cells take up cholesterol by endocytosis.

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Explain how an increase in these LDL receptors will help to reduce blood cholesterol levels.

(2)

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(c) Familial hypercholesterolemia (FH) is a genetic disorder that results in high levels of LDLs in the blood.

(i) In the UK, approximately 270 000 people have FH.

The disorder affects 1 in 250 people in the UK.

Calculate the population size of the UK.

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Express your answer in standard form.

(2)

Answer .....

(ii) The disorder can be caused by a mutation in the gene that codes for LDL receptors.

Suggest why statins are less effective in treating people who are homozygous for this disorder.

(2)

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(iii) Another drug, used to treat FH, is a short antisense strand of DNA from the apo B gene.

Suggest how this antisense strand of DNA would reduce the levels of apo B protein.

(2)

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(Total for question = 11 marks)

Q38.

Meselson and Stahl provided data that supported the theory that DNA replicates by semi-conservative replication.

(a) State the meaning of the term semi-conservative replication.

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(b) Describe the role of DNA polymerase in the process of DNA replication.

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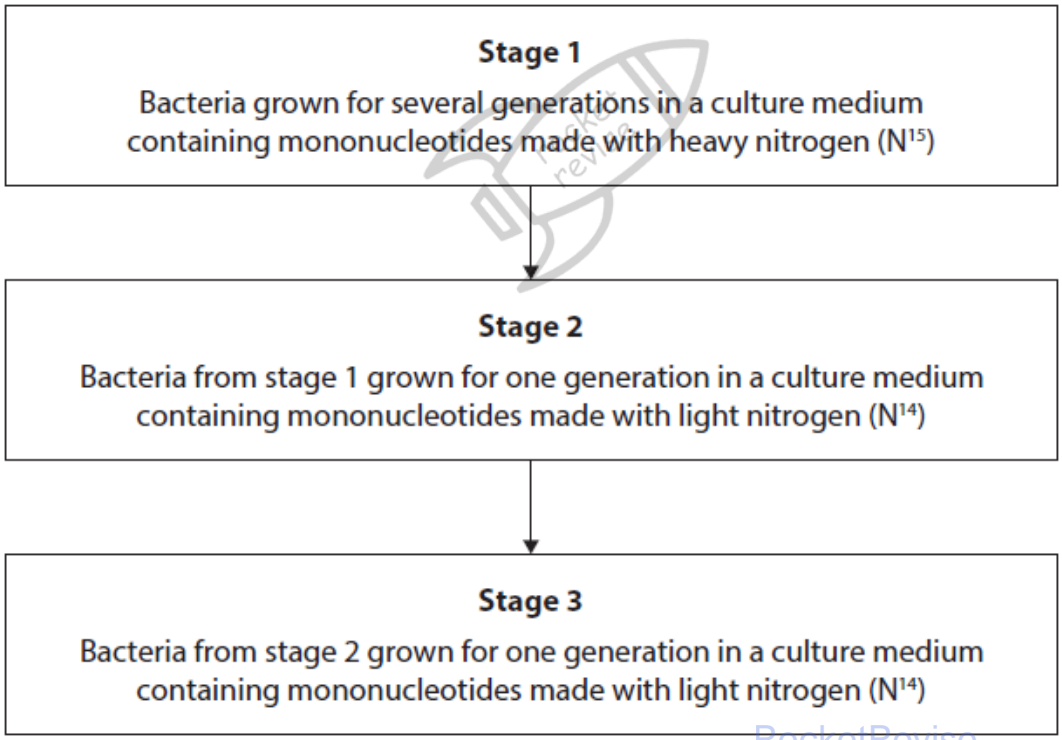
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(c) Meselson and Stahl used mononucleotides made with either heavy nitrogen (N^{15}) or light nitrogen (N^{14}) in their experiments.

The flowchart summarises part of one experiment performed by Meselson and Stahl.



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After each stage, a sample of DNA was taken from the bacteria and the DNA molecules separated on a density gradient in a tube.

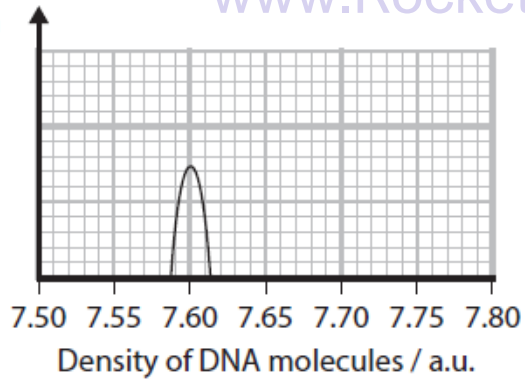
- DNA with only heavy nitrogen has a mean density of 7.74 a.u.
- DNA with only light nitrogen has a mean density of 7.60 a.u.

Meselson and Stahl measured the concentration of DNA with different densities.

The graph shows the results of these measurements for DNA molecules with only light nitrogen.

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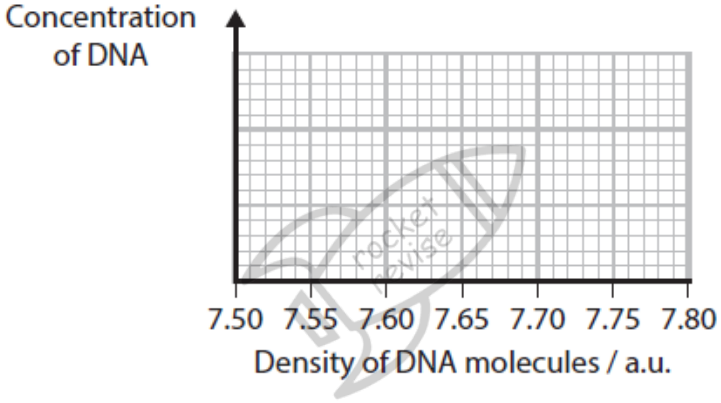


Complete the graphs to show the concentrations and densities of the DNA molecules produced at the end of stages 1, 2 and 3 shown in the flowchart.

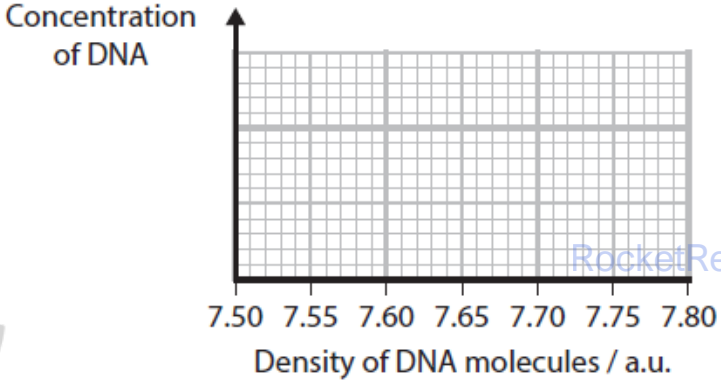
(6)

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End of stage 1:



End of stage 2:

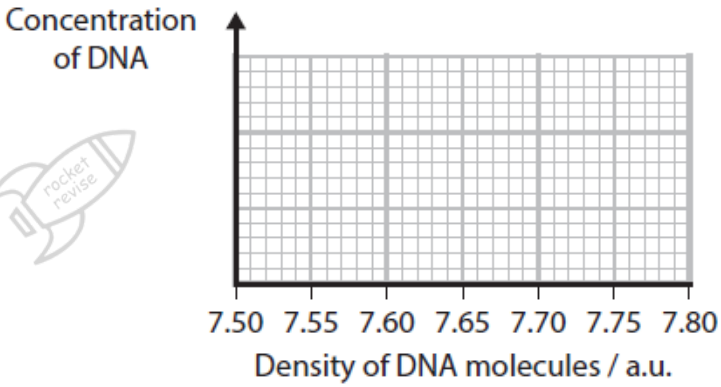


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End of stage 3:



(Total for question = 10 marks)

Q39.

When an organism grows or increases in number, cell division takes place.

(a) When a cell is preparing to divide, DNA synthesis takes place.

Explain how two identical molecules of DNA are produced.

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(b) Protein synthesis increases as a cell grows.

Describe the roles of messenger RNA (mRNA) and transfer RNA (tRNA) in protein synthesis.

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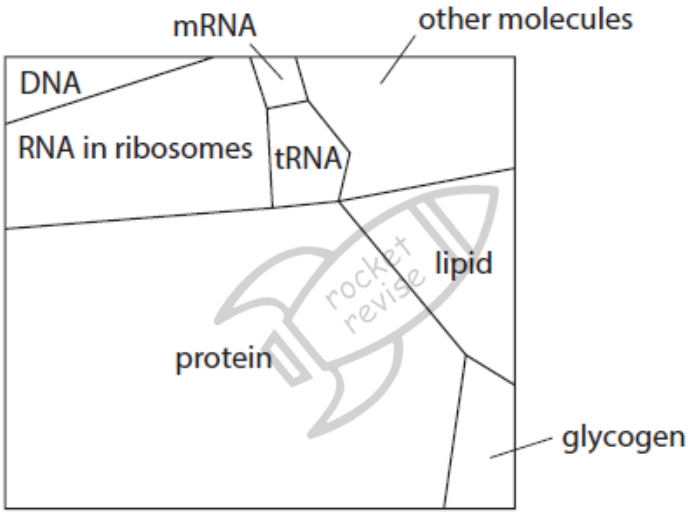
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(c) The Voronoi diagram shows the proportions of molecules in a bacterial cell.

The area of each labelled part of the square is proportional to the percentage of the dry mass of that molecule. The shape of each part has no meaning.

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(Source: http://book.bionumbers.org/what-is-the-macromolecular-composition-of-the-cell/)

(i) Give **one** advantage and **one** disadvantage of using a Voronoi diagram to present this information.

(2)

Advantage

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Disadvantage

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*(ii) The proportions of these molecules change as a bacterial cell prepares to divide, as it divides into two new bacterial cells and as each new bacterial cell grows.

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Discuss how the appearance of this Voronoi diagram might change:

- as a bacterial cell prepares to divide
- in one of the new bacterial cells produced
- as one new bacterial cell increases in size.

Give reasons for these changes to support your answer.

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Q40.

Plastic waste can cause damage to the environment.

Some microorganisms produce enzymes that break down plastic.

Scientists are using protein engineering techniques to modify these enzymes to improve the way in which they break down plastics.

These modifications include:

- changing the properties of the active site
- making enzymes more stable at higher temperatures
- increasing the size of the active site of the enzyme.

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(a) The interaction between enzyme and substrate can be improved by changing the properties of the active site.

Explain what determines the properties of an active site.

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(b) (i) Some types of plastic break down at higher temperatures.
These temperatures are too high for some enzymes to be active.

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Explain why some enzymes cannot function at high temperatures.

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(ii) Changing the bonding in the enzyme can make it more stable at higher temperatures.

Suggest why changing the bonding could enable enzymes to work at higher temperatures.

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(iii) Suggest why changing the bonding could affect the binding of the plastics to the active site of the enzyme.

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(c) A larger active site could affect the activity of the enzyme.

(i) Suggest how a larger active site could **increase the rate** of breakdown of plastic.

(2)

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(ii) Suggest why an enzyme with a larger active site may be **less active.**

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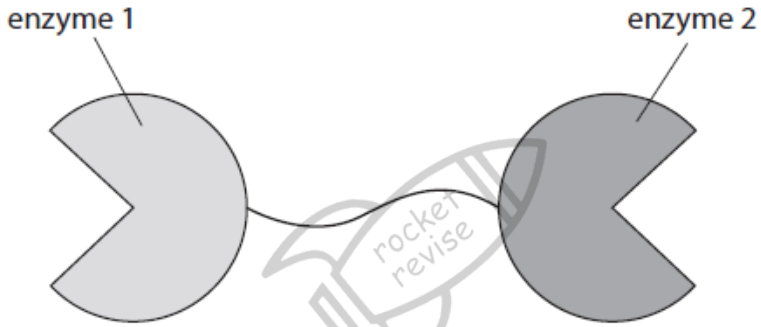
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(d) One way of improving the activity of an enzyme is to link it to a different enzyme.

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The diagram shows two different enzymes linked together.



Enzyme 1 is the enzyme that is engineered by linking it to a different enzyme, enzyme 2.

Enzyme 1 breaks down plastic into short chains of monomers.

Enzyme 2 can break down these short chains into individual monomers.

Explain why linking these two enzymes together will speed up the breakdown of plastics.

(2)

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